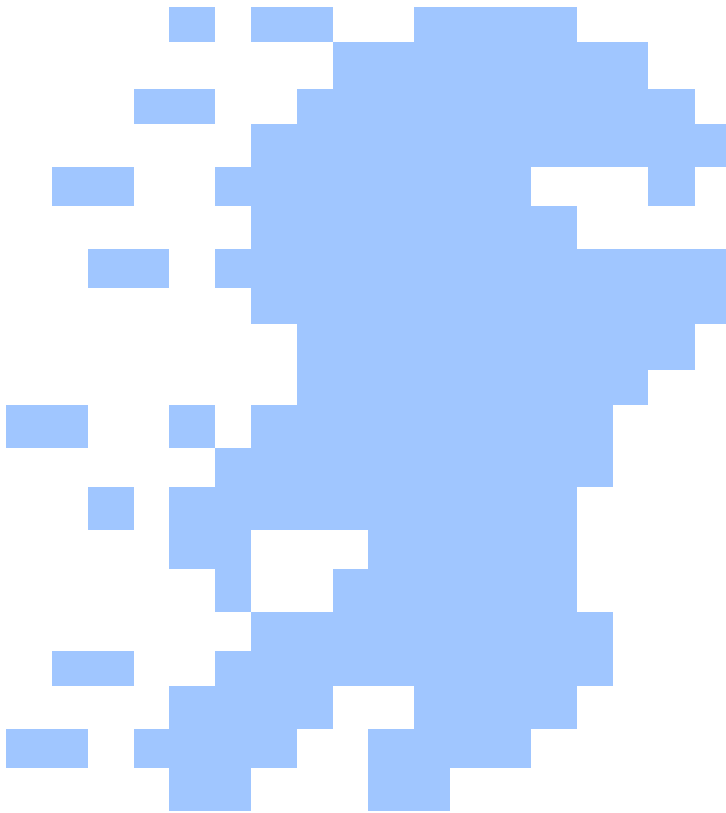


Django Ninja Made Simple

A Beginner-Friendly Guide to Building Modern APIs with Python Django Ninja



Featuring Three Practical Projects &
Step-by-Step Deployment Guide

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Preface

Hey there! You’ve just picked up Django Ninja Made Simple—the fastest, most practical way to get started with Django Ninja. If you’re new to Django or just starting out as a developer, this book’s got your back. You don’t need any experience with Django. I’ll walk you through everything, step by step, from the basics all the way to building a full professional API system. Along the way, you’ll work with real examples, hands-on exercises, and, in the end, you’ll take on a final project: a complete Banking API Capstone. Let’s dive in and start building!

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Chapter 1

Introduction to Django Ninja

Building APIs the Easy Way

What Is Django Ninja?

Django Ninja is a tool that helps you build APIs (Application Programming Interfaces) using Python and Django — but faster, cleaner, and with less code.

It's called “***Ninja***” because it's:

- **Fast:** (built on top of *FastAPI* ideas)
- **Simple:** (you write fewer lines of code)
- **Smart:** (it checks your data automatically)

So instead of writing a lot of code to make a simple API, Django Ninja does most of the heavy lifting for you.

Why Should You Learn Django Ninja?

Because it makes API development easy — especially if you are a beginner in backend development or Django.

Here's what makes it special:

Feature	Django Ninja	Django REST Framework (DRF)
Speed	Very fast (built on ASGI)	Slower (WSGI)
Code size	Less code	More boilerplate
Validation	Uses Pydantic automatically	Needs manual setup
Docs	Auto-generates Swagger & ReDoc	Optional setup
Learning curve	Easy for beginners	Steeper learning

What Django Ninja Does for You

When you create an API, Django Ninja:

- Takes your incoming data (from a web or mobile app)
- Checks it automatically (Is it valid? Is it complete?)
- Passes it to your function to process

- Returns a clean JSON response
- And also creates interactive API documentation for you to test right away
- All that — from just a few lines of code!

Example: Your First Django Ninja API

Here's how easy it is to make your first API with Django Ninja:

```
# api.py

from ninja import NinjaAPI

api = NinjaAPI()

@api.get("/hello")
def say_hello(request):
    return {"message": "Hello, Django Ninja!"}
```

Let's Understand What's Happening:

NinjaAPI() → This is your main API app (like a container for your endpoints).

@api.get("/hello") → This tells Django Ninja to make an API endpoint at /hello.

say_hello() → The function that runs when someone visits /hello.

It returns a dictionary, and Django Ninja automatically converts it to JSON.

Now, run your Django project and open:

<http://127.0.0.1:8000/api/docs>

You'll see a beautiful Swagger documentation page where you can test your /hello endpoint — no setup needed.

The “API Docs” Magic

The `/api/docs` page is a live testing playground that Django Ninja builds automatically for you.

It shows:

- Every endpoint in your project
- The required inputs
- Expected outputs
- And a “Try it out” button to test directly from your browser
- This saves hours of manual testing.

Beginner Tip

Django Ninja is not a replacement for Django, it works with Django. You still use Django models, ORM, and settings, but now your APIs are easier to manage.

Key Components (You’ll Learn These Later)

Component	Purpose
NinjaAPI	The main API app
Routers	Group related endpoints together
Schemas (Pydantic Models)	Define input/output data
Endpoints	Functions that respond to API calls
Docs	Built-in testing & documentation (Swagger / ReDoc)

Real-World Uses of Django Ninja

You can use Django Ninja to build:

- Backend for mobile apps (e.g., Flutter, React Native)
- APIs for web dashboards (e.g., Vue, React)
- Internal data services in organizations
- AI model endpoints (for ML predictions)
- Banking or E-commerce systems
- It’s flexible and production-ready.

Quick Analogy

Think of Django Ninja like a friendly translator between your website/app and your backend logic.

User → (Request) → Django Ninja → (Response) → User

You just tell Django Ninja *what* to send and *where*, and it takes care of the rest.

Common Terms You'll Hear Often

Term	Meaning
API	A way for software to communicate
Endpoint	A specific path (like /hello) in your API
Request	Data coming into your API
Response	Data going out from your API
JSON	The data format used to send and receive info
Swagger	A web tool that shows your API docs automatically

Exercises

1. What is Django Ninja in your own words?
2. What's the difference between Django Ninja and Django REST Framework?
3. Create a simple API endpoint /greet that returns your name.
4. Open your /api/docs page and test it.

Chapter 2

Setting Up Your Development Environment

Preparing Your System to Build Django Ninja APIs

The Goal of This Chapter

Before you start building APIs, you need a **proper setup** — your tools, editor, and Python environment.

In this chapter, you'll learn how to:

- Install Python and Django Ninja
- Create a virtual environment
- Set up your first Django project
- Verify that everything works correctly

Step 1: Check If Python Is Installed

Open your **terminal** (Command Prompt, PowerShell, or VS Code Terminal) and type:

- `python --version`
- You should see something like:
- Python 3.10.11

Django Ninja works best with **Python 3.9 or later**.

If you don't have Python installed:

1. Go to <https://python.org/downloads>
2. Download the latest Python version for your OS.
3. During installation, check **“Add Python to PATH”**.

Step 2: Create a Project Folder

Choose a folder where you'll keep all your code projects.

For example:

➤ `mkdir django_ninja_project`

- `cd django_ninja_project`
- This will be your working directory.

Step 3: Create a Virtual Environment

A **virtual environment** keeps your project's Python packages separate from your computer's system packages.

Create one with:

```
python -m venv venv
```

Then activate it:

- On **Windows**:
 - `venv\Scripts\activate`
- On **Mac/Linux**:
 - `source venv/bin/activate`

Once activated, you'll see `(venv)` appear at the start of your terminal line — that means your virtual environment is active.

Beginner Tip:

Always activate your virtual environment before running or installing anything related to Django.

Step 4: Install Django and Django Ninja

Now install both **Django** and **Django Ninja** using `pip`:

```
pip install django django-ninja
```

Wait a few seconds. Once done, you can verify the installations:

```
python -m django --version
```

You should see something like:

5.1.2

and Django Ninja should now be available as part of your installed packages.

Step 5: Create a Django Project

Now, let's create your main Django project.

Run this:

```
django-admin startproject ninja_api .
```

That will create several files and folders:

ninja_api/

├── __init__.py

├── settings.py

├── urls.py

├── asgi.py

└── wsgi.py

These are the core files of any Django project.

You'll mainly work with **settings.py**, **urls.py**, and **manage.py**.

Step 6: Run the Development Server

Now test that your Django project runs correctly:

```
python manage.py runserver
```

If everything is okay, you'll see something like:

Starting development server at <http://127.0.0.1:8000/>

Now open that link in your web browser.

You should see the **Django welcome page!**

Step 7: Connect Django Ninja

Now, let's connect Django Ninja to your Django project.

Create a new file named **api.py** in the same folder as **manage.py**.

Inside it, paste this:

```
# api.py

from ninja import NinjaAPI

api = NinjaAPI()

@api.get("/hello")
def hello(request):

    return {"message": "Hello from Django Ninja!"}
```

Then, open your `ninja_api/urls.py` file and add this:

```
from django.contrib import admin

from django.urls import path

from api import api

urlpatterns = [

    path('admin/', admin.site.urls),

    path('api/', api.urls),

]
```

Now, run your server again:

```
python manage.py runserver
```

Go to:

<http://127.0.0.1:8000/api/docs>

You'll see the **Django Ninja Swagger documentation** — ready to test your first endpoint

Beginner Tips

- Always activate your virtual environment before running commands.

- Keep your api.py clean and organized — it's where your endpoints live.
- Visit /api/docs to test APIs without writing frontend code.
- If the server fails to start, check if api is imported correctly in urls.py.

Troubleshooting

Issue	Fix
'django' is not recognized	Reinstall Python or check PATH
'ModuleNotFoundError: No module named ninja'	Run pip install django-ninja again
Server not loading	Make sure you are in the correct folder (manage.py directory)

Exercise

Try these steps:

1. Create your own API endpoint /welcome that returns your name.
2. Test it on the Swagger docs page.
3. Create a new endpoint /add that takes two numbers and returns their sum.

Example:

```
@api.get("/add")
```

```
def add(request, a: int, b: int):
```

```
    return {"result": a + b}
```

Visit /api/docs and try it out!

Chapter 3

Your First Ninja API Project

Building Your First Real API Step by Step

The Goal of This Chapter

Now that you have Django and Django Ninja installed and running, it's time to **build your first real API**.

You'll learn:

- How API routes (endpoints) work
- How to organize your code
- How to accept user input
- How to return data as JSON

By the end of this chapter, you'll have a working **mini API project**.

What Is an Endpoint?

An **endpoint** is simply a URL path your API listens to.

For example:

Endpoint	Description
/hello	Returns a greeting
/add	Adds two numbers
/users	Returns a list of users

When someone visits your endpoint (via a web or mobile app), Django Ninja runs the Python function linked to that endpoint.

The Basic Structure of a Ninja API

You already created a simple `api.py` in Chapter 2.

Here's how a typical `api.py` looks:

```
# api.py  
  
from ninja import NinjaAPI
```

```
api = NinjaAPI()

@api.get("/hello")

def hello(request):

    return {"message": "Hello, Django Ninja!"}
```

When you open <http://127.0.0.1:8000/api/docs>, Django Ninja automatically shows this `/hello` endpoint in the documentation — ready to test.

Creating Multiple Endpoints

You can create as many endpoints as you want, just by adding new decorators like `@api.get()` or `@api.post()`.

Let's add a few examples:

```
# api.py

from ninja import NinjaAPI

api = NinjaAPI()

@api.get("/hello")

def hello(request):

    return {"message": "Hello, Django Ninja!"}

@api.get("/greet")

def greet_user(request, name: str):

    return {"greeting": f"Hello, {name}!"}

@api.get("/add")

def add_numbers(request, a: int, b: int):
```

```
return {"result": a + b}
```

What's Happening Here?

- `/greet?name=John` → returns “Hello, John!”
- `/add?a=10&b=20` → returns `{ "result": 30 }`

Django Ninja automatically converts query parameters like `a=10` into Python integers (`int`).

Different Request Types (GET, POST, PUT, DELETE)

APIs use **HTTP methods** to define what action to perform:

Method	Action	Example
GET	Retrieve data	<code>/users</code>
POST	Create new data	<code>/users</code>
PUT	Update existing data	<code>/users/1</code>
DELETE	Remove data	<code>/users/1</code>

You'll use these to create **CRUD** operations (Create, Read, Update, Delete) later.

Example of a POST endpoint:

```
@api.post("/welcome")
def welcome_user(request, name: str):
    return {"message": f"Welcome, {name}!"}
```

When you test this from Swagger, it will send a POST request with name in the form data or JSON.

Handling URL Path Parameters

Sometimes you want to include a value **inside the URL**, not as a query.

Example:

```
@api.get("/user/{user_id}")
```

```
def get_user(request, user_id: int):  
    return {"user_id": user_id, "status": "Found"}
```

If you visit `/api/user/10`, Django Ninja passes 10 to the function automatically.

Beginner Tip

Query parameters (`?a=1&b=2`) are best for filtering or optional data.
Path parameters (`/user/10`) are for identifying specific items.

Organizing Your Endpoints (Routers)

As your project grows, your `api.py` file can get big.
That's where **Routers** come in — they help organize related endpoints.

Example:

```
# users.py  
  
from ninja import Router  
  
router = Router()  
  
@router.get("/list")  
def list_users(request):  
    return {"users": ["Alice", "Bob", "Charlie"]}
```

Now in `api.py`:

```
from ninja import NinjaAPI  
  
from users import router as users_router  
  
api = NinjaAPI()  
api.add_router("/users", users_router)
```

Now your docs will show:

- `/users/list` → returns a list of users.

You can create routers for different modules (e.g., users, products, orders) to keep code clean.

Testing It All

Start your server:

```
python manage.py runserver
```

Visit:

<http://127.0.0.1:8000/api/docs>

You'll see all endpoints listed — `/hello`, `/greet`, `/add`, `/users/list`, etc. Click on any one, hit “Try it out”, and you'll get instant results!

Common Mistakes (and Fixes)

Mistake	Fix
Forgot to add <code>api.urls</code> in <code>urls.py</code>	Add <code>path("api/", api.urls)</code>
Missing type hints (<code>a: int</code>, <code>b: int</code>)	Always include data types
Browser shows “Not Found”	Check the endpoint path carefully
Error: “import not found”	Make sure the file name and import path are correct

Exercises

1. Create an endpoint `/multiply` that takes `x` and `y` and returns their product.
2. Add a new router for products with a `/list` endpoint that returns a few product names.
3. Create a `/welcome/{name}` endpoint that returns `"Welcome, <name>!"`.

Chapter 4

Request & Response Models (Schemas)

Teaching Your API How to Speak in Organized Data

The Goal of This Chapter

You've built simple endpoints using strings and numbers. Now it's time to teach your API to **handle real-world data** like user info, orders, or posts — in a structured, reliable way.

We'll do that using **Schemas**, which are like data blueprints.

By the end of this chapter, you'll know:

- What schemas are
- How to send and receive structured data
- How Django Ninja automatically validates and documents your data

What Is a Schema?

A **Schema** is a Python class that describes the shape of your data.

For example, if you're building a user registration API, your data might look like this:

```
{  
  "username": "JohnDoe",  
  "email": "john@example.com",  
  "age": 25  
}
```

To represent that data in Django Ninja, you create a Schema using **Pydantic** (a powerful data validation library built into Django Ninja).

Creating a Schema

Here's how to define one:

```
from ninja import Schema

class UserSchema(Schema):

    username: str

    email: str

    age: int
```

That's it!

This class tells Django Ninja what to expect — and it will automatically:

- Validate incoming data
- Convert data to JSON
- Generate API documentation fields

Using Schema for Input (Request Body)

Now, let's accept user data in an endpoint:

```
from ninja import NinjaAPI, Schema

api = NinjaAPI()

class UserSchema(Schema):

    username: str

    email: str

    age: int

@api.post("/register")
def register_user(request, data: UserSchema):

    return {"message": f"User {data.username} registered successfully!"}
```

How It Works:

- When a user sends JSON data (username, email, age),
- Django Ninja automatically converts it into a UserSchema object (data),
- You can then access data.username, data.email, etc.

Try it in Swagger!

It automatically shows the form fields and checks the data types for you.

Sending Structured Data Back (Response Model)

Now let's return a schema as the response too.

```
class UserResponse(Schema):  
  
    id: int  
  
    username: str  
  
    email: str  
  
@api.post("/create", response=UserResponse)  
def create_user(request, data: UserSchema):  
  
    return {"id": 1, "username": data.username, "email": data.email}
```

Now, Django Ninja automatically formats and documents the response structure.

Behind the Scenes

When you add response=UserResponse, Ninja:

- Validates your output
- Converts Python objects to JSON
- Documents the result in Swagger

This makes your API **predictable and consistent**.

Example: Full Create and Get User Flow

```
from ninja import NinjaAPI, Schema
```

```
api = NinjaAPI()

class UserIn(Schema):
    username: str
    email: str

class UserOut(Schema):
    id: int
    username: str
    email: str

users = [] # pretend database

@api.post("/users", response=UserOut)
def create_user(request, data: UserIn):
    user_id = len(users) + 1
    new_user = {"id": user_id, **data.dict()}
    users.append(new_user)
    return new_user

@api.get("/users", response=list[UserOut])
def list_users(request):
    return users
```

What's Happening:

- /users (POST) → Creates a user
- /users (GET) → Lists all created users

Try it in Swagger — you'll see both routes, full form fields, and automatic documentation.

Data Validation Magic

Django Ninja automatically checks input data for you.

For example, if your schema says:

```
class Product(Schema):
```

```
    name: str
```

```
    price: float
```

Then sending this:

```
{ "name": "Laptop", "price": "abc" }
```

will return:

```
{ "detail": [{ "loc": ["body", "price"], "msg": "value is not a valid float" }] }
```

No manual validation needed!

Default Values and Optional Fields

You can set defaults and make some fields optional.

```
from typing import Optional
```

```
class Book(Schema):
```

```
    title: str
```

```
    author: Optional[str] = "Unknown"
```

```
    year: int = 2024
```

If a user omits author, it'll automatically become "Unknown".

Quick Tips

- Use **Input Schema** for data coming in (like UserIn)
- Use **Output Schema** for data going out (like UserOut)
- Always use **type hints** (str, int, float, bool, etc.)
- Swagger auto-generates docs for all fields

Try It Yourself

1. Create a schema called ProductIn with fields: name (str), price (float), stock (int)
2. Create ProductOut with an extra field id (int)
3. Build:
 - /products (POST) → adds a product
 - /products (GET) → lists all products

Chapter 5

Connecting Django Ninja with a Database (Models + ORM)

Teaching Your API to Remember Things

The Goal of This Chapter

So far, your APIs can accept and send data — but every time you restart the server, all that data disappears.

It's time to give your API a **memory** by connecting it to a **database** using Django's ORM (Object Relational Mapper).

By the end of this chapter, you'll be able to:

- Connect Django Ninja to a database
- Create and use models
- Save, retrieve, and list data using Django ORM

What is an ORM?

ORM stands for **Object Relational Mapper**.

It allows you to interact with your database using **Python objects** instead of raw SQL.

For example:

```
# Instead of SQL like this

INSERT INTO users (username, email) VALUES ('John', 'john@example.com');


# You can do this

user = User(username="John", email="john@example.com")

user.save()
```

ORMs make your code **cleaner**, **safer**, and **database-independent**.

Step 1: Setting Up Your Database

Django comes preconfigured with **SQLite** — perfect for beginners. It stores data in a single file (like `db.sqlite3`) and requires no setup.

If you open your project folder, you'll see this line in `settings.py`:

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.sqlite3',  
        'NAME': BASE_DIR / "db.sqlite3",  
    }  
}
```

You can keep this as is for now.

Step 2: Creating a Model

A **model** defines a table in your database. Let's create one for our users.

Create a file inside your Django app (e.g. `users/models.py`):

```
from django.db import models  
  
class User(models.Model):  
    username = models.CharField(max_length=50)  
    email = models.EmailField(unique=True)  
    age = models.IntegerField()  
  
    def __str__(self):  
        return self.username
```

Each field represents a column in the database table.

Step 3: Run Migrations

Once the model is ready, tell Django to create the actual table in the database:

```
python manage.py makemigrations
```

```
python manage.py migrate
```

Done!

You now have a `users_user` table in your database.

Step 4: Connecting the Model to Ninja API

Now, let's create endpoints that use our User model.

Inside `api.py` (or your main views file):

```
from ninja import NinjaAPI, Schema
```

```
from .models import User
```

```
api = NinjaAPI()
```

```
class UserIn(Schema):
```

```
    username: str
```

```
    email: str
```

```
    age: int
```

```
class UserOut(Schema):
```

```
    id: int
```

```
    username: str
```

```
email: str

age: int


@api.post("/users", response=UserOut)
def create_user(request, data: UserIn):

    user = User.objects.create(**data.dict())

    return user


@api.get("/users", response=list[UserOut])
def list_users(request):

    return list(User.objects.all())
```

What's Happening Here

- `User.objects.create(**data.dict())` creates and saves a new record.
- `User.objects.all()` fetches all users from the database.
- The schemas make sure the input and output are properly structured.

Step 5: Viewing Your Data

After creating a few users in Swagger, check your database.

You can open the Django shell:

```
python manage.py shell

Then run:

from users.models import User

User.objects.all()
```

You'll see your data stored — persistence achieved!

Example Output

POST /users

```
{  
  "username": "Alice",  
  "email": "alice@example.com",  
  "age": 28  
}
```

Response

```
{  
  "id": 1,  
  "username": "Alice",  
  "email": "alice@example.com",  
  "age": 28  
}
```

GET /users

```
[  
  {  
    "id": 1,  
    "username": "Alice",  
    "email": "alice@example.com",  
    "age": 28  
  }  
]
```

Updating and Deleting Data (Preview)

We'll cover this properly in the next chapter (CRUD Operations), but here's a sneak peek:

```
@api.put("/users/{user_id}", response=UserOut)

def update_user(request, user_id: int, data: UserIn):

    user = User.objects.get(id=user_id)

    for attr, value in data.dict().items():

        setattr(user, attr, value)

    user.save()

    return user


@api.delete("/users/{user_id}")

def delete_user(request, user_id: int):

    User.objects.filter(id=user_id).delete()

    return {"message": "User deleted successfully!"}
```

Try It Yourself

1. Create a model for Product with fields: name, price, and stock.
2. Create schemas (ProductIn, ProductOut) for API I/O.
3. Build /products (POST) and /products (GET) endpoints.
4. Test in Swagger UI and check the database.

Chapter 6

CRUD Operations in Django Ninja

Create, Read, Update, Delete — the Four Pillars of Every API

The Goal of This Chapter

Every useful API needs to do four basic things:

1. **Create** data
2. **Read** data
3. **Update** data
4. **Delete** data

These are called **CRUD operations**, and Django Ninja makes them *super easy* to build.

By the end of this chapter, you'll have a fully functional CRUD API that works with a database and validates everything automatically.

What is CRUD?

CRUD stands for:

Operation	HTTP Method	Description
Create	POST	Add new data
Read	GET	Retrieve data
Update	PUT or PATCH	Modify existing data
Delete	DELETE	Remove data

Let's bring this to life with an example.

Our Example: Managing Products

We'll build a simple Product Management API that can:

- Add new products
- View all products
- Update existing ones
- Delete unwanted ones

Step 1: Define the Model

In `products/models.py`:

```
from django.db import models

class Product(models.Model):

    name = models.CharField(max_length=100)

    price = models.FloatField()

    stock = models.IntegerField(default=0)

    def __str__(self):

        return self.name
```

Run your migrations to create the table:

```
python manage.py makemigrations
python manage.py migrate
```

Step 2: Create Schemas

In `products/api.py`:

```
from ninja import Schema

class ProductIn(Schema):

    name: str

    price: float

    stock: int

class ProductOut(Schema):

    id: int

    name: str
```

```
price: float  
stock: int
```

Step 3: Create the API Endpoints

Now let's implement all CRUD routes using Django Ninja.

```
from ninja import NinjaAPI  
  
from .models import Product  
  
from .api import ProductIn, ProductOut  
  
api = NinjaAPI()  
  
# CREATE  
  
@api.post("/products", response=ProductOut)  
def create_product(request, data: ProductIn):  
    product = Product.objects.create(**data.dict())  
    return product  
  
# READ ALL  
  
@api.get("/products", response=list[ProductOut])  
def list_products(request):  
    return list(Product.objects.all())  
  
# READ SINGLE  
  
@api.get("/products/{product_id}", response=ProductOut)  
def get_product(request, product_id: int):
```

```

    return Product.objects.get(id=product_id)

# UPDATE

@api.put("/products/{product_id}", response=ProductOut)
def update_product(request, product_id: int, data: ProductIn):

    product = Product.objects.get(id=product_id)

    for attr, value in data.dict().items():

        setattr(product, attr, value)

    product.save()

    return product

# DELETE

@api.delete("/products/{product_id}")
def delete_product(request, product_id: int):

    Product.objects.filter(id=product_id).delete()

    return {"message": "Product deleted successfully!"}

```

Understanding the Flow

Action	Endpoint	Description
POST	/products	Create a new product
GET	/products	View all products
GET	/products/{id}	View a single product
PUT	/products/{id}	Update product info

DELETE	/products/{id}	Delete product
---------------	----------------	----------------

You can test all these actions directly in **Swagger UI** (automatically generated at /api/docs).

Example Interaction

POST /products

```
{  
  "name": "Laptop",  
  "price": 899.99,  
  "stock": 10  
}
```

Response

```
{  
  "id": 1,  
  "name": "Laptop",  
  "price": 899.99,  
  "stock": 10  
}
```

GET /products

```
[  
  {  
    "id": 1,  
    "name": "Laptop",  
    "price": 899.99,  
    "stock": 10  
  }  
]
```

Common Errors (and Fixes)

Error: Product.DoesNotExist

Happens when you try to access a product that doesn't exist.

Fix:

```
from django.shortcuts import get_object_or_404

@api.get("/products/{product_id}", response=ProductOut)
def get_product(request, product_id: int):
    product = get_object_or_404(Product, id=product_id)
    return product
```

Error: Missing field in POST

Django Ninja automatically checks for missing or invalid fields — and returns a helpful validation error.

Example:

```
{
  "detail": [
    {
      "loc": ["body", "price"],
      "msg": "field required"
    }
  ]
}
```

No manual checks needed — Ninja and Pydantic do the hard work!

Optional Improvement: PATCH for Partial Updates

Sometimes you don't want to update *all* fields — only one or two. That's what PATCH is for.

```
from typing import Optional

class ProductPatch(Schema):
    name: Optional[str]
    price: Optional[float]
    stock: Optional[int]

@api.patch("/products/{product_id}", response=ProductOut)
def partial_update(request, product_id: int, data: ProductPatch):
    product = Product.objects.get(id=product_id)
    for attr, value in data.dict(exclude_unset=True).items():
        setattr(product, attr, value)
    product.save()
    return product
```

Now you can send only the fields you want to change.

Chapter 7

Authentication Basics

Keeping Your API Safe — Only the Right People Get In

The Goal of This Chapter

Up to now, your API accepts and returns data freely — which is great for testing, but terrible for security.

We need to make sure only **authorized users** can access certain endpoints. This is where **authentication** comes in.

By the end of this chapter, you'll understand:

- What authentication is
- How to protect endpoints in Django Ninja
- How to use Django's built-in authentication system

What Is Authentication?

Authentication means confirming *who* someone is.

When you log in to a website or mobile app, you enter your credentials — usually an email/username and password. If correct, you're allowed to access protected resources.

Example:

- Anyone can *view* a public product list (no login).
- Only admins can *add* or *delete* products (login required).

Step 1: Using Django's Built-in Auth System

Django already comes with a full authentication system — including:

- Users
- Passwords (securely hashed)
- Sessions (for logged-in users)
- Permissions & Groups

We'll build on top of that using Django Ninja.

Step 2: Add Django's Auth App

Open your settings.py and ensure this line is present:

```
INSTALLED_APPS = [  
    ...,  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
]
```

Run migrations if you haven't already:

```
python manage.py migrate
```

This creates the user tables in your database.

Step 3: Create a User Registration API

Let's allow new users to sign up via the API.

```
from django.contrib.auth.models import User  
from ninja import NinjaAPI, Schema  
  
api = NinjaAPI()  
  
class RegisterIn(Schema):  
    username: str  
    email: str  
    password: str  
  
class RegisterOut(Schema):  
    id: int
```

```

username: str

email: str

@api.post("/register", response=RegisterOut)
def register(request, data: RegisterIn):

    user = User.objects.create_user(

        username=data.username,

        email=data.email,

        password=data.password

    )

    return user

```

This securely saves the user with a hashed password.

Step 4: Basic Authentication (Login Required)

Django Ninja supports **Basic Authentication** — a simple username + password check for endpoints.

Add this to api.py:

```

from ninja.security import HttpBasicAuth

class BasicAuth(HttpBasicAuth):

    def authenticate(self, request, username, password):

        from django.contrib.auth import authenticate

        user = authenticate(username=username, password=password)

        if user:

            return user

```

Now use it to protect an endpoint:

```

auth = BasicAuth()

@api.get("/profile", auth=auth)
def get_profile(request):
    user = request.auth # authenticated user object
    return {"message": f"Welcome {user.username}!"}

```

How It Works

1. When calling /profile, the client must include credentials (username & password).
2. Django verifies them using its built-in user system.
3. If valid → access granted.
4. If invalid → 401 Unauthorized.

Swagger UI even provides a small **"Authorize"** button so you can test authentication easily.

Step 5: Restricting Access to Admins Only

You can add simple permission logic:

```

@api.get("/admin-only", auth=auth)
def admin_area(request):
    user = request.auth
    if not user.is_staff:
        return api.create_response(request, {"error": "Access denied"}, status=403)
    return {"message": f"Welcome, admin {user.username}!"}

```

This ensures only staff members (users with `is_staff=True`) can access this route.

Step 6: Testing the Authentication Flow

1. Use /register to create a new user.
2. Go to /profile in Swagger.
3. Click “Authorize” → enter your username and password.
4. Test again — you should see your welcome message.

Success! Your first protected API endpoint works.

Bonus: Using Django's Built-in Login System

You can also use Django's built-in login/logout logic if you want to integrate with web sessions:

```
from django.contrib.auth import login, logout

@api.post("/login")
def login_user(request, data: RegisterIn):
    from django.contrib.auth import authenticate
    user = authenticate(username=data.username, password=data.password)
    if user:
        login(request, user)
        return {"message": "Login successful"}
    return {"error": "Invalid credentials"}

@api.post("/logout")
def logout_user(request):
    logout(request)
    return {"message": "Logged out successfully"}
```

This works with Django session-based authentication (great for web apps).

Try It Yourself

1. Create /register and /profile endpoints.
2. Use BasicAuth to protect /profile.
3. Test logging in with valid and invalid credentials.
4. Add a new /admin-only route for staff users.

Chapter 8

JWT Authentication

Modern, Token-Based Security for Your Django Ninja API

The Goal of This Chapter

Basic Authentication is simple, but it's not ideal for production apps — credentials are sent with every request.

Modern systems use **JWT (JSON Web Tokens)** — short-lived, secure tokens that clients can store and send automatically.

By the end of this chapter, you'll know:

- What JWT is and why it's used
- How to set up JWT authentication in Django Ninja
- How to protect routes using tokens

What Is JWT?

A **JWT (JSON Web Token)** is a secure digital token that identifies a user. Instead of sending a username/password every time, a user logs in once and gets a **token**.

That token is then sent with every request.

Typical flow:

1. User logs in → API gives a JWT token.
2. User includes token in headers → API checks it → Grants access.

JWTs are **stateless** — they don't require Django sessions or cookies. They're ideal for mobile and modern web apps.

Step 1: Install the JWT Library

Django Ninja doesn't include JWT by default, but it works perfectly with the popular library **PyJWT**.

Install it:

```
pip install PyJWT
```

Step 2: Create a JWT Utility File

In your app folder, create `auth_utils.py`:

```
import jwt

from datetime import datetime, timedelta

from django.conf import settings

# Secret key from settings.py
JWT_SECRET = settings.SECRET_KEY
JWT_ALGORITHM = "HS256"

def create_jwt_token(user):
    payload = {
        "user_id": user.id,
        "username": user.username,
        "exp": datetime.utcnow() + timedelta(hours=1) # expires in 1 hour
    }
    return jwt.encode(payload, JWT_SECRET, algorithm=JWT_ALGORITHM)

def decode_jwt_token(token):
    try:
        payload = jwt.decode(token, JWT_SECRET, algorithms=[JWT_ALGORITHM])
        return payload
    except jwt.ExpiredSignatureError:
```

```
        return None

    except jwt.InvalidTokenError:

        return None
```

This file creates and validates JWTs.

Step 3: Create Login API That Returns JWT

```
from django.contrib.auth import authenticate

from ninja import NinjaAPI, Schema

from .auth_utils import create_jwt_token


api = NinjaAPI()

class LoginIn(Schema):

    username: str

    password: str

class LoginOut(Schema):

    access_token: str

    token_type: str = "Bearer"

@api.post("/login", response=LoginOut)

def login(request, data: LoginIn):

    user = authenticate(username=data.username, password=data.password)

    if not user:

        return api.create_response(request, {"error": "Invalid credentials"}, status=401)

    token = create_jwt_token(user)
```

```
return {"access_token": token}
```

When a user logs in, they receive a token like:

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

Step 4: Create JWT Authentication Class

In `api.py`:

```
from ninja.security import HttpBearer
from .auth_utils import decode_jwt_token
from django.contrib.auth.models import User

class JWTAuth(HttpBearer):

    def authenticate(self, request, token):
        payload = decode_jwt_token(token)
        if payload:
            try:
                user = User.objects.get(id=payload["user_id"])
                return user
            except User.DoesNotExist:
                return None
```

Now this `JWTAuth` can be attached to any endpoint you want to secure.

Step 5: Protect an Endpoint

```
auth = JWTAuth()

@api.get("/profile", auth=auth)
```

```
def get_profile(request):
    user = request.auth
    return {"username": user.username, "email": user.email}
```

Now, to access `/profile`, the user must send a valid JWT token.

Swagger UI automatically provides a **“Authorize”** button.

Enter your token as:

Bearer <your_token_here>

Example Flow

Login

POST /login

```
{
  "username": "alice",
  "password": "password123"
}
```

Response

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer"
}
```

Access Protected Route

GET /profile

Header:

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

Response

```
{
  "username": "alice",
```

```
"email": "alice@example.com"
}
```

Step 6: Handling Expired Tokens

If the token expires (after 1 hour in our example), the API will respond:

```
{"detail": "Unauthorized"}
```

You can refresh tokens by simply re-logging in or building a `/refresh-token` route.

Optional: Token Expiry and Refresh

If you want users to get a **new token before the old one expires**, you can add:

```
def refresh_jwt_token(old_token):
    payload = decode_jwt_token(old_token)
    if payload:
        user_id = payload["user_id"]
        username = payload["username"]
        new_payload = {
            "user_id": user_id,
            "username": username,
            "exp": datetime.utcnow() + timedelta(hours=1)
        }
        return jwt.encode(new_payload, JWT_SECRET, algorithm=JWT_ALGORITHM)
    return None
```

Try It Yourself

1. Register a user.
2. Log in to get a JWT token.
3. Copy the token and paste it in Swagger's **Authorize** section.
4. Try accessing `/profile` (it will work only if token is valid).

Chapter 9

Role-Based Permissions

Controlling What Users Can Access in Your API

The Goal of This Chapter

Now that we've learned how to authenticate users with JWT, the next step is **authorization** — deciding *what a user can do* after they log in.

By the end of this chapter, you'll learn:

- The difference between **authentication** and **authorization**
- How to assign user roles (e.g., admin, editor, viewer)
- How to restrict routes based on user permissions
- How to make your API smarter and more secure

Authentication vs. Authorization

Concept	Description	Example
Authentication	Confirms <i>who</i> the user is	Logging in with username/password
Authorization	Determines <i>what</i> the user can do	Admin can delete users; viewer cannot

They work together:

1. **Authentication** → Validates the user (using JWT).
2. **Authorization** → Checks their permissions (using roles or groups).

Step 1: Add Roles to Users

There are two common ways to manage roles:

1. Use Django's built-in **Groups** and **Permissions**
2. Create a **custom field** in your user model (simpler for small apps)

Let's use the **custom field** method for clarity.

Update models.py

```
from django.contrib.auth.models import AbstractUser

from django.db import models

class User(AbstractUser):

    ROLE_CHOICES = (

        ("admin", "Admin"),

        ("editor", "Editor"),

        ("viewer", "Viewer"),

    )

    role = models.CharField(max_length=10, choices=ROLE_CHOICES, default="viewer")


    def __str__(self):

        return f'{self.username} ({self.role})'
```

Then run:

```
python manage.py makemigrations
```

```
python manage.py migrate
```

Step 2: Create Sample Users with Roles

You can create users using Django Admin or Python shell:

```
python manage.py createsuperuser
```

Or directly in shell:

```
from app.models import User

User.objects.create_user(username="alice", password="1234", role="admin")

User.objects.create_user(username="bob", password="1234", role="editor")

User.objects.create_user(username="eve", password="1234", role="viewer")
```

Step 3: Create Role-Based Permission Decorator

We'll create a simple helper function to check if a user has the right role.

Create a new file: permissions.py

```
from ninja.errors import HttpError

def role_required(allowed_roles):

    def decorator(func):

        def wrapper(request, *args, **kwargs):

            user = request.auth

            if user.role not in allowed_roles:

                raise HttpError(403, f"Permission denied: {user.role} cannot access this resource.")

            return func(request, *args, **kwargs)

        return wrapper

    return decorator
```

Now you can easily protect routes by specifying roles.

Step 4: Use Role-Based Decorators on Routes

In your api.py:

```
from .permissions import role_required

from .auth_utils import JWTAuth

auth = JWTAuth()

@api.get("/admin/users", auth=auth)
```

```

@role_required(["admin"])

def list_users(request):

    from django.contrib.auth.models import User

    users = User.objects.all().values("id", "username", "email", "role")

    return list(users)


@api.post("/edit/article", auth=auth)
@role_required(["admin", "editor"])
def edit_article(request):

    return {"message": f"{request.auth.username} edited the article!"}


@api.get("/read/article", auth=auth)
@role_required(["admin", "editor", "viewer"])
def read_article(request):

    return {"message": "You can view this article"}

```

Example Flow:

- alice (admin) → can list users, edit, and read.
- bob (editor) → can edit and read.
- eve (viewer) → can only read.

Step 5: Test It in Swagger

1. Log in as bob (editor).
2. Copy your JWT token.
3. Click **Authorize** in Swagger → Paste Bearer <token>.
4. Try to call /admin/users → You'll get a **403 Forbidden**.
5. Try /edit/article → Works
6. Try /read/article → Works

This proves your permission system works correctly.

Optional: Use Django's Built-in Groups

If you're building a larger system, you can use:

```
from django.contrib.auth.models import Group
```

Each group (like “Admin”, “Editor”, etc.) can be linked to permissions dynamically.

But for learning and small apps, our custom field method is simpler and easier to understand.

Exercise

1. Add a new role called "moderator".
2. Give moderators the ability to read and edit, but not delete.
3. Update the decorator to allow this new role in /edit/article.
4. Test it by creating a moderator user and checking your API routes.

Chapter 10

API Versioning and Documentation Mastery

Keeping Your APIs Organized, Compatible, and Well-Explained

The Goal of This Chapter

As your API grows, new versions may be needed — for bug fixes, feature upgrades, or compatibility reasons. You'll also want clear, automatic documentation for users and developers.

By the end of this chapter, you'll learn:

- How to organize and version your APIs (v1, v2, etc.)
- How to customize Django Ninja's **Swagger / ReDoc** docs
- How to provide rich documentation with examples and tags

Step 1: What Is API Versioning?

Imagine your mobile app calls your API. You update it and change some routes — suddenly, old apps stop working

Versioning solves this.

It lets you publish new versions while keeping old ones running.

Example:

```
/api/v1/users
```

```
/api/v2/users
```

Both can exist at the same time, and clients choose which version to use.

Step 2: Implementing Versioning in Django Ninja

In Django Ninja, versioning is super easy — just create multiple NinjaAPI instances.

Example:

```
from ninja import NinjaAPI

api_v1 = NinjaAPI(version="1.0.0", title="Banking API - v1")

api_v2 = NinjaAPI(version="2.0.0", title="Banking API - v2")
```

You can register both in your project's `urls.py`:

```
from django.urls import path

from .api_v1 import api as api_v1
from .api_v2 import api as api_v2

urlpatterns = [

    path("api/v1/", api_v1.urls),
    path("api/v2/", api_v2.urls),
]
```

Now, you can update your routes independently without breaking older clients.

Step 3: Organizing Endpoints by Version

Let's imagine v1 returns basic user info, and v2 adds account balance.

api_v1.py

```
from ninja import NinjaAPI

api = NinjaAPI(title="Banking API v1")

@api.get("/users")
def list_users(request):
    return [{"id": 1, "name": "Alice"}]
```

api_v2.py

```
from ninja import NinjaAPI

api = NinjaAPI(title="Banking API v2")
```

```
@api.get("/users")

def list_users(request):

    return [{"id": 1, "name": "Alice", "balance": 2000}]
```

Now, visiting:

- /api/v1/docs → Version 1 docs
- /api/v2/docs → Version 2 docs

Both versions coexist safely.

Step 4: Auto Documentation (Swagger & ReDoc)

Django Ninja automatically generates:

- **Swagger UI** (interactive testing)
- **ReDoc UI** (clean reference-style documentation)

Visit:

<http://127.0.0.1:8000/api/v1/docs>

or

<http://127.0.0.1:8000/api/v1/redoc>

You'll see all routes, models, and example requests.
No extra setup required!

Step 5: Customizing Documentation

You can enhance the docs using parameters inside route decorators.

Example:

```
from ninja import Router, Schema

router = Router(tags=["User Management"])

class UserOut(Schema):
```

```

id: int

name: str

email: str

@router.get("/users", response=list[UserOut], summary="Get all users",
description="Returns a list of all users in the system.")
def get_users(request):

    return [

        {"id": 1, "name": "Alice", "email": "alice@example.com"},

        {"id": 2, "name": "Bob", "email": "bob@example.com"},

    ]

```

The docs will show:

- **Tag:** “User Management”
- **Summary:** Short title
- **Description:** Full explanation

This makes your docs look professional and easy to navigate.

Step 6: Adding Examples and Metadata

You can show **request/response examples** in Swagger for better clarity.

```

class CreateUserIn(Schema):

    name: str

    email: str

class CreateUserOut(Schema):

    id: int

    name: str

```

```

email: str

@router.post(
    "/users",
    response=CreateUserOut,
    summary="Create new user",
    description="Registers a new user and returns their details",
    examples={
        "input": {"name": "John", "email": "john@example.com"},
        "output": {"id": 1, "name": "John", "email": "john@example.com"}
    }
)

def create_user(request, data: CreateUserIn):
    return {"id": 1, **data.dict()}

```

Swagger will show “Try it out” examples — perfect for beginners testing your API.

Step 7: Customizing Global API Info

When creating your NinjaAPI instance, you can personalize the documentation details:

```

api = NinjaAPI(
    title="Cyber Oracle Banking API",
    version="2.0.0",
    description="A secure and high-performance banking backend powered by Django Ninja.",
    docs_url="/docs",
    openapi_url="/openapi.json"
)

```

Your docs will now show your title, version, and description beautifully.

Exercise

1. Create a v1 and v2 version of your “Banking API.”
2. In v2, add a new field called balance to the User model output.
3. Add Swagger tags and a summary to each route.
4. Visit both `/v1/docs` and `/v2/docs` — compare the results.

Chapter 11

Error Handling & Validation

Making Your API Smart, Reliable, and Foolproof

The Goal of This Chapter

Your API should never crash or confuse users when bad data or errors occur. This chapter teaches you how to gracefully handle such issues.

By the end, you'll learn:

- How Django Ninja automatically validates input
- How to define custom error messages
- How to handle server and client errors
- How to send clean, structured responses for all situations

Step 1: Built-in Validation with Pydantic

Django Ninja uses **Pydantic** under the hood — a library that validates data automatically based on Python type hints.

Example:

```
from ninja import Schema, NinjaAPI

api = NinjaAPI()

class UserIn(Schema):
    name: str
    age: int

@api.post("/users")
def create_user(request, data: UserIn):
    return {"message": f"User {data.name} ({data.age}) created successfully"}
```

Now, if a client sends:

```
{"name": "Alice", "age": "twenty"}
```

Ninja instantly returns:

```
{  
  "detail": [  
    {  
      "type": "int_parsing",  
      "loc": ["body", "age"],  
      "msg": "Input should be a valid integer"  
    }  
  ]  
}
```

No manual validation needed — Ninja handles it for you!

Step 2: Adding Default Values and Constraints

You can add validation rules and defaults inside your schemas.

```
from pydantic import Field  
  
class RegisterUser(Schema):  
    username: str = Field(..., min_length=3, max_length=20)  
    password: str = Field(..., min_length=6)  
    email: str = Field(..., regex=r"^[\\w\\.]+@[\\w\\.]+\\.\\w+$")
```

Now, Ninja will:

- Reject usernames shorter than 3 characters
- Reject invalid emails
- Require all fields (... means required)

Step 3: Handling Common Errors Gracefully

If something goes wrong in your route, you can raise a **HttpError**.

```
from ninja.errors import HttpError

@api.get("/divide")
def divide(request, a: int, b: int):
    if b == 0:
        raise HttpError(400, "Division by zero is not allowed.")

    return {"result": a / b}
```

Example:

GET /divide?a=10&b=0 →

Response:

```
{
  "detail": "Division by zero is not allowed."
}
```

Step 4: Custom Error Responses

You can also define your own error format using **exception handlers**.

```
from ninja.errors import HttpError
from django.http import JsonResponse

@api.exception_handler(HttpError)
def custom_http_error(request, exc):
    return JsonResponse({
        "success": False,
        "error": str(exc),
        "path": request.path
    })
```

```
}, status=exc.status_code)
```

Now all errors will return in a neat, uniform JSON structure:

```
{
  "success": false,
  "error": "Division by zero is not allowed.",
  "path": "/divide"
}
```

Step 5: Global Exception Handling

Sometimes an unexpected server error might occur. You can catch *all* exceptions using `Exception` in a handler.

```
@api.exception_handler(Exception)
def global_exception_handler(request, exc):
    return JsonResponse({
        "success": False,
        "error": "Unexpected server error. Please try again later."
    }, status=500)
```

This ensures users never see ugly error traces.

Step 6: Validation for Query & Path Parameters

You can also validate query parameters using type hints:

```
@api.get("/search")
def search_users(request, keyword: str, limit: int = 10):
    if limit > 100:
        raise HttpError(400, "Limit cannot exceed 100.")
```

```
return {"message": f"Searching for {keyword}, limit {limit}"}
```

- keyword is required
- limit defaults to 10
- Raises an error if limit > 100

Step 7: Custom Validation with Pydantic Validators

If you want to apply complex rules, use `@validator`.

```
from pydantic import validator

class ProductIn(Schema):
    name: str
    price: float

    @validator("price")
    def price_must_be_positive(cls, v):
        if v <= 0:
            raise ValueError("Price must be greater than zero")

        return v
```

Invalid input:

```
{"name": "Laptop", "price": -100}
```

Response:

```
{
  "detail": [
    {
      "loc": ["body", "price"],
```

```
"msg": "Price must be greater than zero",  
"type": "value_error"  
}  
]  
}
```

Exercise

1. Add validation for user registration (name, email, password).
2. Add a /calculate route that validates positive numbers only.
3. Add a global handler for all ValueError exceptions.
4. Test it using Swagger's "Try It Out" feature.

Chapter 12

Pagination & Filtering

Keeping API Responses Fast, Clean, and Easy to Navigate

The Goal of This Chapter

As your app grows, you'll likely have **hundreds or thousands of records** — users, products, transactions, etc.

Returning them all at once would be **slow** and **wasteful**. That's where **pagination and filtering** come in.

By the end of this chapter, you'll learn:

- How to use Django Ninja's built-in pagination
- How to create custom pagination logic
- How to filter and search through your database
- How to combine both effectively

Step 1: Why Pagination?

Without pagination:

GET /users

→ returns 10,000 users

With pagination:

GET /users?limit=10&offset=20

→ returns 10 users (from position 20)

Much faster. Much cleaner.

Step 2: Django Ninja's Built-in Pagination

Django Ninja provides ready-to-use pagination utilities.

Example:

```
from ninja import NinjaAPI, Schema

from ninja.pagination import paginate, PageNumberPagination

api = NinjaAPI()

class UserOut(Schema):
    id: int
    name: str
    email: str

users_db = [
    {"id": i, "name": f"User {i}", "email": f"user{i}@example.com"}
    for i in range(1, 101)
]

@api.get("/users", response=list[UserOut])
@paginate(PageNumberPagination)
def list_users(request):
    return users_db
```

Now when you visit `/users?page=1`, `/users?page=2`, etc., you'll see 10 results per page (default limit = 10).

Step 3: Customize Pagination Settings

You can create your own pagination class:

```
from ninja.pagination import PageNumberPagination

class CustomPagination(PageNumberPagination):
```

```
page_size = 5
max_page_size = 50
```

Then use it:

```
@api.get("/users", response=list[UserOut])
@paginate(CustomPagination)
def list_users(request):
    return users_db
```

Now each page shows only 5 users.

Step 4: Offset Pagination

Offset pagination is another style:
Instead of pages, it uses numeric offsets.

```
from ninja.pagination import LimitOffsetPagination

@api.get("/users-offset", response=list[UserOut])
@api.paginate(LimitOffsetPagination)
def list_users_offset(request):
    return users_db
```

Now you can query:

/users-offset?limit=5&offset=10

Step 5: Adding Filtering to Your API

Filtering lets users search or narrow down results.

Example:

```
from typing import Optional

@api.get("/search-users", response=list[UserOut])
```

```
def search_users(request, name: Optional[str] = None):
    if name:
        return [u for u in users_db if name.lower() in u["name"].lower()]
    return users_db
```

Now:

```
/search-users?name=User 3
```

returns:

```
[{"id": 3, "name": "User 3", "email": "user3@example.com"}]
```

Step 6: Combine Pagination + Filtering

Both can work together perfectly!

```
@api.get("/users/filter", response=list[UserOut])
@paginate(PageNumberPagination)
def filter_users(request, name: Optional[str] = None):
    data = users_db
    if name:
        data = [u for u in users_db if name.lower() in u["name"].lower()]
    return data
```

Example:

```
/users/filter?name=User&page=2
```

returns paginated, filtered results!

Step 7: Database Filtering (ORM Way)

When using Django models, filtering becomes even easier.

```
from app.models import User
```

```
@api.get("/users-db", response=list[UserOut])
@paginate(PageNumberPagination)
def list_users_from_db(request, name: Optional[str] = None):
    users = User.objects.all()
    if name:
        users = users.filter(name__icontains=name)
    return users
```

Exercise

1. Create an endpoint `/transactions` that lists all transactions with pagination.
2. Add a query parameter type (e.g., deposit, withdrawal) for filtering.
3. Combine both pagination and filtering.
4. Test it in Swagger and try different page numbers.

Chapter 13

Middleware & CORS

Controlling and Securing Every Request to Your API

The Goal of This Chapter

Every request to your API goes through several steps — preprocessing, authentication, logging, etc.

Middleware gives you full control over this process.

By the end of the chapter, you'll know:

- What middleware is and how it works in Django Ninja
- How to write custom middleware
- How to enable CORS to allow safe access from the frontend/mobile apps
- How to use third-party CORS libraries

Step 1: What is Middleware?

Middleware is a **layer** that processes requests before they reach your API endpoints — and responses before they go back to the client.

Think of it as a *security guard* or *filter* at the gate.

In Django:

- Middleware is global (applies to all routes)
- You add middleware in the settings file

Step 2: Creating Your First Custom Middleware

Let's track how long each API request takes.

Step 1: Create a file (e.g. `middleware.py`) in your app:

```
import time
```

```
class RequestTimingMiddleware:

    def __init__(self, get_response):

        self.get_response = get_response

    def __call__(self, request):

        start_time = time.time()

        response = self.get_response(request)

        duration = time.time() - start_time

        print(f"Request to {request.path} took {duration:.4f}s")

        return response
```

Step 2: Add it to your settings

```
MIDDLEWARE = [

    'django.middleware.security.SecurityMiddleware',

    # ...

    'your_app.middleware.RequestTimingMiddleware',

]
```

Now, every request is logged with how long it took.

Step 3: Adding Authorization Globally

You can reject certain requests before they hit any view.
Example: Block all requests from unauthorized IPs.

```
class BlockSpecificIPMiddleware:

    def __init__(self, get_response):

        self.get_response = get_response

        self.blocked_ips = ["192.168.0.100"]
```

```
def __call__(self, request):
    if request.META.get('REMOTE_ADDR') in self.blocked_ips:
        from django.http import JsonResponse
        return JsonResponse({"error": "Not allowed"}, status=403)
    return self.get_response(request)
```

Step 4: What is CORS & Why You Need It

CORS = **Cross-Origin Resource Sharing**

By default, browsers block requests from websites hosted at different domains or ports.

Example:

Frontend:

http://localhost:3000

Backend API: http://localhost:8000

Requests will get blocked unless CORS is enabled.

Step 5: Install & Enable Django CORS Headers

Install:

```
pip install django-cors-headers
```

Add to settings.py:

```
INSTALLED_APPS = [
    ...
    'corsheaders',
    ...
]

MIDDLEWARE = [
    'corsheaders.middleware.CorsMiddleware',
    ...
]
```

Allow all origins (for development):

```
CORS_ALLOW_ALL_ORIGINS = True
```

Step 6: Allow Only Specific Origins (Recommended in Production)

```
CORS_ALLOWED_ORIGINS = [  
    "https://yourfrontend.com",  
    "https://another-allowed-client.com",  
]
```

Step 7: Allow Custom Headers & Methods

```
CORS_ALLOW_HEADERS = [  
    "content-type",  
    "authorization",  
    "x-custom-header",  
]  
  
CORS_ALLOW_METHODS = [  
    "GET",  
    "POST",  
    "PUT",  
    "PATCH",  
    "DELETE"  
]
```

Now only those methods & headers are allowed.

Exercise

1. Add a middleware to log every API request path & method
2. Enable CORS and allow only your frontend app
3. Try making frontend requests before and after adding CORS

Chapter 14

Background Tasks

Running Time-Consuming Work Without Making Users Wait

The Goal of This Chapter

Not all tasks should run while the user waits for a response.

Imagine API tasks like:

- Sending emails
- Processing payment
- Generating reports
- Sending push notifications

These can take time — and can slow down your API responses.

This is where **background tasks** come in. They let you process work **after** sending a response, so your API stays fast.

Step 1: What Are Background Tasks?

A background task is a **function that runs after your main API response is returned**.

The user doesn't wait for it to finish.

Django Ninja supports background tasks through its integration with Starlette.

Step 2: Adding a Basic Background Task

Step 1: Import BackgroundTasks:

```
from ninja import NinjaAPI, BackgroundTasks

api = NinjaAPI()
```

Step 2: Define a background task:

```
def send_welcome_email(email: str):

    print(f'Sending welcome email to {email}...')
```

```
# Simulate delay

import time

time.sleep(3)

print("Email sent!")
```

Step 3: Add it to your endpoint:

```
@api.post("/register")

def register_user(request, email: str, background_tasks: BackgroundTasks):

    background_tasks.add_task(send_welcome_email, email)

    return {"message": f"Registration successful for {email}"}
```

What Happens?

When someone calls:

```
POST /register?email=test@example.com
```

The response returns **immediately**:

```
{
  "message": "Registration successful for test@example.com"
}
```

Meanwhile, the email is sent **in the background**.

No lag for the user.

Step 3: Background Tasks With Arguments

You can pass multiple arguments:

```
def generate_report(user_id: int, include_pdf: bool = False):

    print(f"Generating report for user {user_id}...")

    if include_pdf:
```

```
print("PDF version included.")

print("Report ready!")

Use it like this:

@api.post("/report")

def generate_user_report(request, user_id: int, background_tasks: BackgroundTasks):

    background_tasks.add_task(generate_report, user_id, True)

    return {"message": "Report generation started"}
```

Step 4: Background Tasks with Database

Yes, you can even use Django ORM inside background tasks.

```
def notify_user(user_id: int):

    from app.models import User

    user = User.objects.get(id=user_id)

    print(f'Notifying {user.email}')
```

Use normally: `background_tasks.add_task(notify_user, new_user.id)`

Warning: Background Tasks Are Not Persistent

By default, background tasks run **in-memory**.

If you restart your server, pending tasks will be lost. For serious workloads (like sending massive emails), use **Celery** + Redis.

Optional: Using Celery for Advanced Background Processing

If your application needs distributed, persistent job handling, use Celery. (Main workflow involves a task queue, worker processes, and a broker like Redis).

We'll cover Celery deeper in the Appendix.

Exercise

1. Add a background task to:
 - Send a welcome SMS
 - Generate a user statistics file
2. Simulate delay using `time.sleep()`
3. Watch the task run asynchronously in the console while the response returns immediately.

Chapter 15

WebSockets Support

Adding Real-Time, Two-Way Communication to Your Ninja API

The Goal of This Chapter

Some applications need **instant, live updates** — think chat apps, notifications, live dashboards, or multiplayer games.

That's where **WebSockets** come in — they allow **two-way communication** between your server and client in real-time.

By the end of this chapter, you'll know:

- How to enable WebSockets support in Django Ninja
- How to manage connections
- How to send and receive messages
- Real-world use cases for live APIs

Step 1: What are WebSockets?

Unlike normal HTTP requests (one-way: client → server), WebSockets allow continuous **two-way communication**.

Good for:

- Live chat apps
- Live feeds or notifications
- Multiplayer games
- IoT updates
- Stock or crypto price trackers

Step 2: WebSocket Support in Django Ninja

Django Ninja supports WebSockets via **NinjaAPI.ws_route**.

And because it's built on ASGI, it's super-fast for real-time apps.

Step 3: Your First WebSocket Route

```
from ninja import NinjaAPI

api = NinjaAPI()

@api.ws_route("/ws/echo")
async def echo(request, ws):

    await ws.accept()

    while True:

        message = await ws.receive_text()

        await ws.send_text(f"Echo: {message}")
```

What this does:

- Client connects to `/ws/echo`
- Server accepts connection
- Any message sent by the client is **echoed back**

Step 4: Testing Your WebSocket

You can test this using:

- A frontend JS WebSocket client
- Browser dev console
- Tools like Postman (WebSocket tab)
- VS Code REST client

Example (Quick Test in Browser Console):

```
let ws = new WebSocket('ws://localhost:8000/ws/echo')

ws.onopen = () => ws.send('Hello Server')

ws.onmessage = (msg) => console.log(msg.data)
```

Expected Output:

Echo: Hello Server

Step 5: Sending JSON Instead of Text

```
@api.ws_route("/ws/json")

async def json_test(request, ws):

    await ws.accept()

    await ws.send_json({"message": "Connected!"})

    while True:

        data = await ws.receive_json()

        await ws.send_json({"you_sent": data})
```

When client sends:

```
{"name": "Alice"}
```

Response is:

```
{"you_sent": {"name": "Alice"}}
```

Step 6: Broadcasting to Multiple Clients

To build something like a live chat, you can store connections:

```
connections = []

@api.ws_route("/ws/broadcast")

async def broadcast(request, ws):

    await ws.accept()

    connections.append(ws)

    try:

        while True:

            message = await ws.receive_text()

            for conn in connections:
```

```

        await conn.send_text(f"Broadcast: {message}")

    except:

        connections.remove(ws)

```

Result:

- All connected clients receive the same message
- Works like a room chat

Important Notes

- WebSockets require **async** functions
- They **stay open** until the client/server closes it
- Use proper cleanup to avoid memory leaks

Real-World Use Cases

Use Case	How WebSockets Help
Chat App	Live message updates
Notifications	Instant alerts on UI
Live Dashboard	Push real-time numbers
IoT	Device status updates
Games	Live moves across players

Exercise

1. Create a WebSocket endpoint that counts the number of connected users
2. Build a simple "real-time notification" WebSocket that sends a timestamp every 5 seconds
3. Test using browser console or Postman

Chapter 16

API Documentation & OpenAPI

Automatically Document Your API Like a Pro — No Extra Work!

The Goal of This Chapter

- API What API documentation is and why it matters
- How Django Ninja generates interactive docs
- How to customize the docs with metadata
- How you can share and version the docs easily

documentation helps both developers and clients understand how to consume your API — and Django Ninja automatically provides this out of the box using **OpenAPI** and **Swagger UI**.

By the end of this chapter, you'll understand:

Step 1: What is OpenAPI?

The **OpenAPI Specification** is a standard format for describing REST APIs.

Django Ninja uses OpenAPI to generate:

- Machine-readable API specs
- Human-readable interactive docs (via Swagger UI or Redoc)

Step 2: Automatic Documentation

As soon as you define your API using Ninja:

```
from ninja import NinjaAPI

api = NinjaAPI()

@api.get("/hello")
def hello(request):
```

```
return {"message": "Hello, Ninja!"}
```

You AUTOMATICALLY get API documentation at:

- /docs
- /openapi.json
- /redoc

Example:

```
http://localhost:8000/api/docs
```

Try it now — you'll see your hello endpoint clearly displayed!

Step 3: Swagger UI (Interactive Docs)

The /docs route uses **Swagger UI**, which allows you to:

- See every endpoint
- Try requests live
- View request/response models
- Explore the API structure

It's **interactive, easy to explore, and client-friendly**.

Step 4: Customizing Your API Metadata

You can customize the title, version, and description of your API:

```
api = NinjaAPI(  
    title="My Awesome API",  
    version="1.0.0",  
    description="This API does amazing things using Django Ninja!"  
)
```

It shows up at the top of the API docs

Step 5: Adding Authentication to Docs

If you're using authentication, Django Ninja includes it automatically in the docs.

Example using Bearer Tokens:

```
from ninja.security import HttpBearer

class Auth(HttpBearer):

    def authenticate(self, request, token):

        if token == "supersecret":

            return token

api = NinjaAPI(auth=Auth())
```

The docs will now show an **Authorize** button for adding the token.

Step 6: Dependency Documentation

If you're using dependencies or security, Django Ninja includes them in the docs:

```
from ninja import Depends

def get_token(request):

    token = request.headers.get("X-TOKEN")

    if token != "secret":

        raise Exception("Invalid token")

    return token

@api.get("/secure-data")

def secure_data(request, token=Depends(get_token)):

    return {"data": "Top secret"}
```

The doc will show:

- Required headers
- Security dependencies

Step 7: Exporting OpenAPI Spec

You can share the full API documentation spec as JSON:

- Visit: `/openapi.json`

This helps with:

- API client generation
- Documentation syncing
- Sharing specs with other teams

Bonus: ReDoc UI

Want a different look for your API docs?

Django Ninja also supports ReDoc at `/redoc`.

It's more static and suitable for public-facing docs.

Exercise

1. Add a custom title and description to your API
2. Add a secure endpoint and test how it shows in the docs
3. Export your OpenAPI spec and view as JSON
4. Try `/redoc` and compare with `/docs`

Chapter 17

CORS, Deployment & Production Setup

Turn your Django Ninja API from Local Project to Live, Secure Service

The Goal of This Chapter

After development and testing, the next step is **Deployment** — taking your API live so others can use it. In this chapter, you'll learn:

- What CORS is and how to enable it
- How to prepare your Django project for production
- Best practices for environment variables and secret keys
- How to deploy to common platforms like Render, Railway, or AWS
- Tips for debugging and maintaining your deployed API

Step 1: What is CORS and Why It Matters

CORS (Cross-Origin Resource Sharing) controls which websites are allowed to make requests to your API.

Example:

- Your backend is at `api.myapp.com`
- Your frontend is at `myfrontend.com`

Without CORS configured, browsers will block requests.

Install CORS Headers for Django

```
pip install django-cors-headers
```

Add to `settings.py`

```
INSTALLED_APPS = [  
    ...  
    "corsheaders",
```

```

]

MIDDLEWARE = [
    "corsheaders.middleware.CorsMiddleware",
    ...
]

CORS_ALLOWED_ORIGINS = [
    "https://example-frontend.com",
    "http://localhost:3000",
]

```

You can also allow all for testing but never do this in production:

```
CORS_ALLOW_ALL_ORIGINS = True
```

Step 2: Security Checklist for Production

Before going live, make sure these are in place:

1. SECRET_KEY

Never hardcode your SECRET_KEY. Use environment variables:

```

import os

SECRET_KEY = os.getenv("SECRET_KEY")

```

Use a .env file or your hosting platform's settings.

2. DEBUG Mode

Turn off debug in production:

```
DEBUG = False
```

3. Allowed Hosts

```
ALLOWED_HOSTS = ["yourdomain.com", "api.yourdomain.com"]
```

4. HTTPS

Make sure all requests are encrypted with HTTPS.

5. Database Configuration

Use a production-grade database like PostgreSQL.

Example for DATABASES:

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': os.getenv("DB_NAME"),
        'USER': os.getenv("DB_USER"),
        'PASSWORD': os.getenv("DB_PASSWORD"),
        'HOST': os.getenv("DB_HOST"),
        'PORT': 5432,
    }
}
```

Step 3: Deployment Options

Here's a quick overview of where you can deploy:

Render.com (Beginner-friendly)

- Supports Django directly
- Automatic deploys from GitHub
- Free tier available (can sleep after inactivity)

Steps:

1. Push your code to GitHub
2. Create a Django web service in Render

3. Add your environment variables
4. Deploy

Railway.app (Easy CI/CD)

- Similar to Render
- Supports PostgreSQL built-in

AWS Elastic Beanstalk (Advanced Level)

- Highly scalable
- Requires extra config for networking, load balancing

Docker + VPS (For professionals)

- Build Docker image
- Use servers like DigitalOcean, Linode
- Full control

Step 4: Deploy with Render

Example: render.yaml

services:

- type: web

name: django-ninja-api

env: python

buildCommand: pip install -r requirements.txt

startCommand: gunicorn project_name.wsgi:application

envVars:

- key: SECRET_KEY

sync: false

Step 5: Final Deployment Checklist

1. CSS & static files collected: `python manage.py collectstatic`
2. Database migrated: `python manage.py migrate`
3. Logs available for monitoring
4. Test API with external tools like Postman

Exercise

1. Install and configure `django-cors-headers`
2. Set up a `.env` file with your `SECRET_KEY`
3. Try deploying your API on **Render.com**
4. Inspect your live `/docs` endpoint

Chapter 18

Bonus Tools, Tips & Tricks for Django Ninja

Accelerate Your Workflow and Master Django Ninja Like a Pro

The Goal of This Chapter

Go beyond the basics! In this chapter, we'll explore:

- Tools to speed up API development
- Auto-generating client SDKs
- Structuring large projects
- API versioning
- Reusability and modular code techniques
- Useful third-party tools for Django Ninja

1. API Versioning (Maintain Multiple Versions)

Versioning allows you to run multiple APIs (v1, v2, etc.) side-by-side for backward compatibility.

```
from ninja import NinjaAPI

api_v1 = NinjaAPI(version="1.0.0", title="My API v1")
api_v2 = NinjaAPI(version="2.0.0", title="My API v2")
```

Your URLs might look like:

```
/api/v1/docs
```

```
/api/v2/docs
```

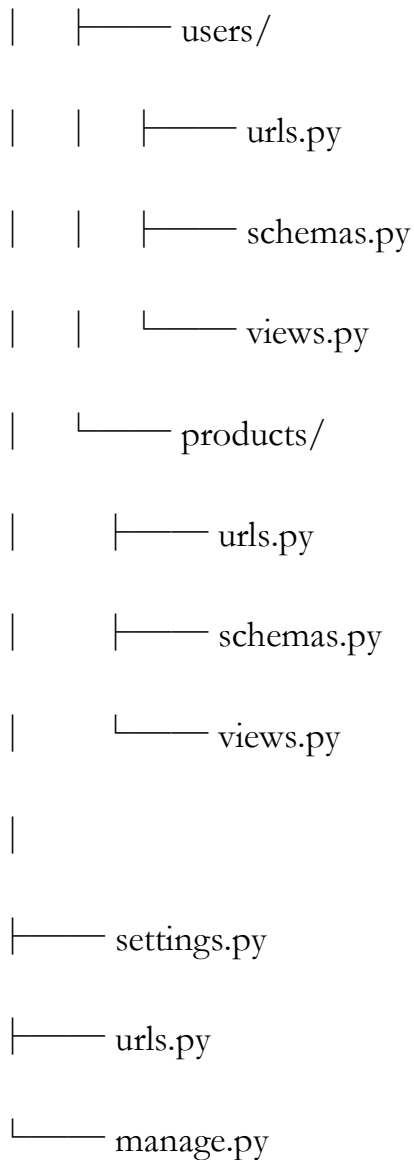
2. Structuring Large Projects

When your API grows, it's important to modularize the code:

Recommended folder structure:

```
project_root/
```

```
├── api/
```



Example: load users router in main API:

```
from api.users.urls import router as users_router

api.add_router("/users", users_router)
```

3. Live API Documentation with ReDoc & Swagger

Ninja automatically provides documentation:

- Swagger UI → <http://localhost:8000/api/docs>
- ReDoc → <http://localhost:8000/api/redoc>

4. Auto-generate API Client SDKs

You can generate client code (for JavaScript, Android, etc.) with tools like:

- **OpenAPI Generator**
- **Swagger Codegen**

```
openapi-generator-cli generate -i http://localhost:8000/api/docs.json -g python -o ./client
```

This helps frontend/mobile teams use your API faster.

5. Useful Third-party Packages

These tools integrate beautifully with Django Ninja:

- **django-ninja-extra**: class-based views, DI
- **django-ninja-jwt**: Simple JWT support
- **drf-yasg**: advanced doc generation
- **django-cors-headers**: CORS management
- **django-stubs**: type hinting support

6. Use .env + Settings Management (Tip!)

Use a library like python-decouple to manage settings cleanly:

```
pip install python-decouple

from decouple import config

SECRET_KEY = config("SECRET_KEY")

DEBUG = config("DEBUG", default=False, cast=bool)
```

Exercise

1. Add a second version of your API using `.add_router`
2. Generate a client SDK using OpenAPI Generator
3. Try reorganizing your app to follow the modular pattern
4. Install and configure `django-ninja-extra` or `django-ninja-jwt`

Chapter 19

Capstone Project – Banking API System

Build a Complete Real-World API Using Django Ninja

The Chapter Goal

In this capstone project, you'll build a fully functional API for a **basic banking system**. This will combine everything we've learned:

- Models & ORM
- CRUD operations
- Authentication
- Permissions & Roles
- Transactions
- Validation & Error handling
- Testing
- Documentation

System Overview

The Banking API will support:

Feature	Description
User Registration	Users can register and login
Account Management	Users can create bank accounts
Deposit & Withdrawal	Transactions affecting account balance
Balance Check	Check account balance anytime
Transaction History	List recent activities
Role-Based Access	Admin vs user permissions

Folder Structure

Here's the folder structure (matching your new graphic):

banking_project/

├── api/

| ├── accounts/

| | ├── schemas.py

| | ├── models.py

| | ├── views.py

| | └── urls.py

| |

| ├── authentication/

| | ├── schemas.py

| | ├── views.py

| | └── urls.py

| |

| └── transactions/

| ├── schemas.py

| ├── models.py

| └── views.py

```
|      └── urls.py
```

```
|
```

```
|── settings.py
```

```
|── urls.py
```

```
└── manage.py
```

Step 1: Create Django Project and Apps

```
django-admin startproject banking_project
```

```
cd banking_project
```

```
python manage.py startapp api
```

Add sub-apps manually:

- accounts
- authentication
- transactions

Step 2: Define Models (accounts/models.py)

```
from django.db import models

from django.contrib.auth.models import User

class Account(models.Model):

    owner = models.OneToOneField(User, on_delete=models.CASCADE)

    account_number = models.CharField(max_length=10, unique=True)

    balance = models.DecimalField(max_digits=12, decimal_places=2, default=0.00)

    created_at = models.DateTimeField(auto_now_add=True)
```

```
def __str__(self):  
    return f'{self.owner.username}'s account"
```

Step 3: Schemas (accounts/schemas.py)

```
from ninja import Schema  
  
class AccountCreate(Schema):  
    account_number: str  
  
class AccountOut(Schema):  
    account_number: str  
    balance: float
```

Step 4: Views & Endpoints (accounts/views.py)

```
from ninja import Router  
from .models import Account  
from .schemas import AccountCreate, AccountOut  
from django.contrib.auth.models import User  
from django.shortcuts import get_object_or_404  
  
router = Router()  
  
@router.post("/create", response=AccountOut)  
def create_account(request, payload: AccountCreate):  
    user = request.user  
    new_account = Account.objects.create(  

```

```

        owner=user,

        account_number=payload.account_number,

        balance=0.00

    )

    return new_account

@router.get("/balance", response=AccountOut)
def get_balance(request):

    account = get_object_or_404(Account, owner=request.user)

    return account

```

Step 5: Add Transaction Logic

Create separate app /transactions that handles:

- Deposit
- Withdrawal
- Listing transactions

Example model (transactions/models.py):

```

class Transaction(models.Model):

    account = models.ForeignKey(Account, on_delete=models.CASCADE)

    amount = models.DecimalField(max_digits=12, decimal_places=2)

    type = models.CharField(max_length=10, choices=[("deposit", "Deposit"), ("withdraw",
"Withdraw")])

    timestamp = models.DateTimeField(auto_now_add=True)

```

Step 6: Authentication & Permissions

Use django-ninja-jwt or your simple session-based approach from earlier chapters to protect endpoints.

Example (protect all account endpoints):

```
from ninja.security import django_auth  
  
router = Router(auth=django_auth)
```

Step 7: Test With Swagger UI

Once you wire everything up in `urls.py` and start the server, visit:

`http://127.0.0.1:8000/api/docs`

You'll interact with your API visually from your browser.

Advanced Features (Optional Challenges)

- Add email notifications on transactions
- Add parallel transfer support
- Use Celery for background tasks
- Add permissions (e.g. staff-only deposits)
- Implement pagination for transaction history

Capstone Exercises

1. Create your own API endpoint to transfer funds between accounts
2. Add admin-only access to an endpoint that lists all users' accounts
3. Write a test script using the Django test client
4. Package and deploy your Banking API on a cloud platform

Chapter 20

Project – Hospital Management System API

A full-featured healthcare backend with Django Ninja

Project Objectives

By the end of this project, you will:

- Build and organize a multi-module API
- Integrate user roles and permissions
- Handle nested relationships (e.g., doctor → appointments → patients)
- Implement CRUD with auth and filtering
- Enable a clean API layer ready for mobile/IoT integration

Features Overview

Module	Features Included
Authentication	Patient vs. Staff vs. Admin roles
Patient Management	Register, update, view medical history
Doctor Management	Create doctor profiles, assign patients
Appointments	Book, update, cancel, check availability
Medical Records	Create, view, and manage a patient's record
Prescriptions	Doctors can prescribe medication to patients
(Optional) Pharmacy	Track drug prescriptions and availability
(Optional) Billing	Generate invoices and track payments

Proposed Folder Structure

hospital_api/

└── api/

| └── auth/

| | └── schemas.py

| | └── views.py

| | └── urls.py

| |

| └── patients/

| | └── schemas.py

| | └── models.py

| | └── views.py

| | └── urls.py

| |

| └── doctors/

| | └── schemas.py

| | └── models.py

| | └── views.py

```

|   |   └── urls.py

|   |

|   └── appointments/

|   |   └── schemas.py

|   |   └── models.py

|   |   └── views.py

|   |   └── urls.py

|   |

|   └── records/

|       └── schemas.py

|       └── models.py

|       └── views.py

|       └── urls.py

|

└── manage.py

└── settings.py

└── urls.py

```

Step-by-Step Workflow

Step 1: Create the Project & Apps

```
django-admin startproject hospital_api  
cd hospital_api  
python manage.py startapp api
```

Inside `api/`, we'll create the sub-apps: `patients`, `doctors`, `appointments`, `records`.

Step 2: Models Example

Let's create a few simplified models to start with.

`api/patients/models.py`

```
from django.db import models  
  
from django.contrib.auth.models import User  
  
class Patient(models.Model):  
    user = models.OneToOneField(User, on_delete=models.CASCADE)  
    dob = models.DateField()  
    address = models.TextField()  
    phone = models.CharField(max_length=20)  
    created_at = models.DateTimeField(auto_now_add=True)
```

`api/doctors/models.py`

```
class Doctor(models.Model):  
    user = models.OneToOneField(User, on_delete=models.CASCADE)  
    specialization = models.CharField(max_length=100)  
    room_number = models.CharField(max_length=10)
```

`api/appointments/models.py`

```

class Appointment(models.Model):

    patient = models.ForeignKey("patients.Patient", on_delete=models.CASCADE)

    doctor = models.ForeignKey("doctors.Doctor", on_delete=models.CASCADE)

    date = models.DateTimeField()

    status = models.CharField(

        max_length=10,

        choices=[("scheduled", "Scheduled"), ("completed", "Completed"), ("cancelled",
"Cancelled")],

        default="scheduled"

    )

    notes = models.TextField(blank=True)

```

APIs With Schemas (Example)

api/appointments/schemas.py

```

from ninja import Schema

from datetime import datetime

from typing import Optional

class AppointmentIn(Schema):

    doctor_id: int

    date: datetime

    notes: Optional[str] = None

class AppointmentOut(Schema):

    id: int

```

```
patient_id: int  
doctor_id: int  
date: datetime  
status: str  
notes: Optional[str]
```

Example Endpoint: Booking an Appointment

api/appointments/views.py

```
from ninja import Router  
from .models import Appointment  
from .schemas import AppointmentIn, AppointmentOut  
from django.shortcuts import get_object_or_404  
from api.patients.models import Patient  
from api.doctors.models import Doctor  
  
router = Router(tags=["appointments"])  
  
@router.post("/", response=AppointmentOut)  
def create_appointment(request, payload: AppointmentIn):  
    patient = get_object_or_404(Patient, user=request.user)  
    doctor = get_object_or_404(Doctor, id=payload.doctor_id)  
    appointment = Appointment.objects.create(  
        patient=patient,  
        doctor=doctor,  
        date=payload.date,
```

```
notes=payload.notes  
  
)  
  
return appointment
```

Security & Roles

You should extend User for both **Patient** and **Doctor** and use Django permissions or a custom role field. Example:

- Patient role: Limited to managing their own records and appointments
- Doctor role: View assigned patients, medical records, create appointments
- Admin role: Full CRUD over all data

Example using Ninja's auth:

```
from ninja.security import django_auth  
  
router = Router(auth=django_auth)
```

Stretch Features to Try

- Add an **admin dashboard endpoint** to see system metrics
- Create a **prescription model** and attach to patient records
- Add a **calendar view** for doctor schedules
- Implement **email or SMS notifications** for appointments
- Add **JWT-based login system**
- Add **pagination and filtering**

Deployment Tips

- Use PostgreSQL for real deployment
- Configure CORS for front-end (React, Angular, mobile apps)
- Set up Nginx + Gunicorn for hosting
- Use Swagger UI (/api/docs) as a client testing layer

Chapter 21

Personal Task Manager API

Build & Manage Tasks with CRUD Operations, Authentication, and Filtering

The Chapter Goal

In this chapter, you will build a fully functional **Task Manager API**, where users can:

- Create accounts
- Login and manage their tasks
- Add, update, delete tasks
- Mark tasks as complete
- Filter tasks (by date, status, or priority)

You'll use everything you've learned so far, including schemas, views, routers, authentication, pagination, and filtering. Let's get started!

API Features Overview

Feature	Description
User Registration	Signup and login to the system
Task CRUD	Create, read, update, delete personal tasks
Status & Priority	Mark tasks as done & assign priority
Filtering	Filter by status, priority, or date
Pagination	List tasks in a paginated format
Documentation	Fully documented via Swagger UI
Security	Token or session authentication

Project Structure

We'll reuse the same modular folder structure as before:

```

task_manager/
├── api/
│   ├── auth/
│   │   ├── schemas.py
│   │   ├── views.py
│   │   └── urls.py
│   └──
├── tasks/
│   ├── schemas.py
│   ├── models.py
│   ├── views.py
│   └── urls.py
└──
    ├── manage.py
    ├── settings.py
    └── urls.py

```

Step 1: Create Django Project & App

```
django-admin startproject task_manager
```

```
cd task_manager  
python manage.py startapp api
```

Step 2: Task Model (api/tasks/models.py)

```
from django.db import models  
from django.contrib.auth.models import User  
  
class Task(models.Model):  
    owner = models.ForeignKey(User, on_delete=models.CASCADE)  
    title = models.CharField(max_length=120)  
    description = models.TextField(blank=True)  
    priority = models.CharField(  
        max_length=10,  
        choices=[("low", "Low"), ("medium", "Medium"), ("high", "High")],  
        default="medium"  
    )  
    status = models.CharField(  
        max_length=10,  
        choices=[("pending", "Pending"), ("done", "Done")],  
        default="pending"  
    )  
    created_at = models.DateTimeField(auto_now_add=True)  
    due_date = models.DateTimeField(null=True, blank=True)  
  
    def __str__(self):
```

```
return self.title
```

Step 3: Schemas for Task API (api/tasks/schemas.py)

```
from ninja import Schema  
from datetime import datetime  
from typing import Optional
```

```
class TaskIn(Schema):  
    title: str  
    description: Optional[str]  
    priority: str = "medium"  
    status: str = "pending"  
    due_date: Optional[datetime]
```

```
class TaskOut(Schema):  
    id: int  
    title: str  
    status: str  
    priority: str  
    created_at: datetime  
    due_date: Optional[datetime]
```

Step 4: Task Views (api/tasks/views.py)

```
from ninja import Router  
from .models import Task  
from .schemas import TaskIn, TaskOut  
from django.shortcuts import get_object_or_404
```

```

router = Router(tags=["tasks"])

@router.post("/", response=TaskOut)
def create_task(request, payload: TaskIn):
    task = Task.objects.create(owner=request.user, **payload.dict())
    return task

@router.get("/", response=list[TaskOut])
def list_tasks(request, status: str = None, priority: str = None):
    tasks = Task.objects.filter(owner=request.user)
    if status:
        tasks = tasks.filter(status=status)
    if priority:
        tasks = tasks.filter(priority=priority)
    return tasks

@router.get("/{task_id}", response=TaskOut)
def get_task(request, task_id: int):
    task = get_object_or_404(Task, owner=request.user, id=task_id)
    return task

@router.put("/{task_id}", response=TaskOut)
def update_task(request, task_id: int, payload: TaskIn):
    task = get_object_or_404(Task, owner=request.user, id=task_id)

```

```
for attr, value in payload.dict().items():
    setattr(task, attr, value)

task.save()

return task

@router.delete("/{task_id}")
def delete_task(request, task_id: int):
    task = get_object_or_404(Task, owner=request.user, id=task_id)

    task.delete()

    return {"success": True}
```

Step 5: Add Authentication

You can reuse your JWT or session-based setup:

```
from ninja.security import django_auth

router = Router(auth=django_auth)
```

Step 6: Add URL Routing (in main urls.py)

```
from django.contrib import admin

from django.urls import path

from ninja import NinjaAPI

from api.tasks.views import router as tasks_router

from api.auth.views import router as auth_router

api = NinjaAPI()

api.add_router("/tasks", tasks_router)

api.add_router("/auth", auth_router)
```

```
urlpatterns = [  
    path("admin/", admin.site.urls),  
    path("api/", api.urls),  
]
```

Testing With Swagger UI

Once the server is running, test your API with all endpoints under:

<http://127.0.0.1:8000/api/docs>

Check:

- Create a task
- List all your tasks
- Update one task
- Delete a task
- Filter by priority or status

Optional Stretch Goals

- Add task reminders via email
- Notify when a task is overdue
- Add search by keyword
- Add user avatars and profiles
- Assign tasks to teammates

Capstone Exercise

1. Build the TaskManager API by following the steps
2. Add at least one **filtering** and one **pagination** feature
3. Secure all task endpoints with authentication
4. Deploy your app and share the live link + Swagger docs
5. Push your code to GitHub as a project portfolio piece

Chapter 22

Deployment & Scaling Django Ninja APIs

Step-by-step to go from local dev → production, and scale safely

What you'll get in this chapter

- Production checklist (quick-read)
- Two primary deployment methods (server + Docker) with full config examples
- Database (Postgres) setup and connection details
- Serving static files (collectstatic) and media files strategy
- Process management (Gunicorn + systemd) + Nginx reverse proxy example
- Docker Compose example (Postgres + Redis + Celery + Nginx + Gunicorn)
- Kubernetes notes (basic manifests for scaling)
- Background tasks (Celery + Redis) and how to run them in production
- Caching (Redis) & connection pooling hints
- Zero-downtime deploys, migrations, health checks
- Security, secrets, logging, backups, monitoring
- Troubleshooting checklist & exercises

Production Checklist (quick)

Before you deploy, ensure:

1. DEBUG = False in production.
2. SECRET_KEY stored in environment variable (never in repo).
3. ALLOWED_HOSTS configured.
4. Database (Postgres) provisioned and reachable.
5. Static files handled (collectstatic) and served by Nginx/CDN.
6. Gunicorn (or ASGI server) configured to run the app.
7. Reverse proxy (Nginx) in front to serve static files & TLS.
8. HTTPS (Let's Encrypt) configured.
9. Logging & monitoring in place.
10. Backups & migrations strategy defined.

11. Health checks and readiness endpoints.
12. Celery workers (if used) and Redis configured.
13. Minimal required ports open, firewall rules enforced.

1. Prepare your Django project for production

1.1 Settings best practices (settings.py)

Move sensitive values to environment variables. Example pattern:

```
# settings.py (fragment)

import os

from pathlib import Path

BASE_DIR = Path(__file__).resolve().parent.parent

SECRET_KEY = os.environ.get("DJANGO_SECRET_KEY")

DEBUG = os.environ.get("DJANGO_DEBUG", "False") == "True"

ALLOWED_HOSTS = os.environ.get("DJANGO_ALLOWED_HOSTS",
                                "localhost").split(",")

# Database via env (Postgres)

DATABASES = {

    "default": {

        "ENGINE": "django.db.backends.postgresql",

        "NAME": os.environ.get("POSTGRES_DB"),

        "USER": os.environ.get("POSTGRES_USER"),

        "PASSWORD": os.environ.get("POSTGRES_PASSWORD"),
```

```

    "HOST": os.environ.get("POSTGRES_HOST", "localhost"),
    "PORT": os.environ.get("POSTGRES_PORT", "5432"),
}
}

# Static files
STATIC_ROOT = BASE_DIR / "staticfiles"
MEDIA_ROOT = BASE_DIR / "mediafiles"

# Security
SECURE_PROXY_SSL_HEADER = ("HTTP_X_FORWARDED_PROTO", "https")
SESSION_COOKIE_SECURE = True
CSRF_COOKIE_SECURE = True

```

Important: Do not commit secret keys or production DB credentials.

2. Database: Postgres setup & config

2.1 Provision a Postgres DB

Use a managed DB (recommended): AWS RDS, DigitalOcean Managed DB, Render DB, Railway Postgres. Or provision on your server with:

Ubuntu example (not for large production)

```

sudo apt update

sudo apt install postgresql postgresql-contrib

sudo -u postgres createuser --interactive # create user

sudo -u postgres createdb mydb

```

Create a database and a role, then set a strong password. Example SQL:

```
CREATE ROLE ninja_user WITH LOGIN PASSWORD 'verystrongpassword';

CREATE DATABASE ninja_db OWNER ninja_user;

GRANT ALL PRIVILEGES ON DATABASE ninja_db TO ninja_user;
```

2.2 Connection string / DATABASE_URL (recommended)

Many platforms use a single DATABASE_URL env var:

```
postgres://USER:PASSWORD@HOST:PORT/DBNAME
```

You can parse this in Django using dj-database-url:

```
# pip install dj-database-url

import dj_database_url

DATABASES['default'] =
dj_database_url.config(default=os.environ.get("DATABASE_URL"))
```

3. Static files and media

Run collectstatic during deploy:

```
python manage.py collectstatic --noinput
```

Serve STATIC_ROOT and MEDIA_ROOT with Nginx (not Django). Optionally push static files to a CDN or S3 for scale.

Nginx snippet (later full example) will map location /static/ to STATIC_ROOT.

4. Option A: Deploy to a single VPS (Gunicorn + Nginx + systemd)

This is classic, understandable, and good for small teams.

4.1 Install system dependencies

(Ubuntu example)

```
sudo apt update

sudo apt install python3-pip python3-venv nginx
```

4.2 Create virtualenv & install requirements

```
cd /srv
```

```

sudo mkdir django_ninja_app

sudo chown $USER:$USER django_ninja_app

cd django_ninja_app

python3 -m venv venv

source venv/bin/activate

pip install -U pip

pip install -r requirements.txt

```

4.3 Gunicorn systemd service

```

Create /etc/systemd/system/gunicorn.service:

[Unit]

Description=gunicorn daemon for django ninja

After=network.target


[Service]

User=www-data

Group=www-data

WorkingDirectory=/srv/django_ninja_app

Environment="DJANGO_SETTINGS_MODULE=project.settings"

Environment="PYTHONUNBUFFERED=1"

Environment="DJANGO_SECRET_KEY=... (set via systemd or env file)"

ExecStart=/srv/django_ninja_app/venv/bin/gunicorn \

    project.wsgi:application \

    --name ninja_app \

    --bind unix:/run/gunicorn.sock \

```

```
--workers 3 \  
  
--threads 2 \  
  
--timeout 120
```

[Install]

```
WantedBy=multi-user.target
```

Reload & start:

```
sudo systemctl daemon-reload  
  
sudo systemctl start gunicorn  
  
sudo systemctl enable gunicorn
```

4.4 Nginx reverse proxy

Create /etc/nginx/sites-available/django_ninja:

```
server {  
  
    listen 80;  
  
    server_name yourdomain.com www.yourdomain.com;  
  
  
    location /static/ {  
  
        alias /srv/django_ninja_app/staticfiles/;  
  
    }  
  
  
    location /media/ {  
  
        alias /srv/django_ninja_app/mediafiles/;  
  
    }  
  
  
    location / {
```

```

    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header Host $http_host;
    proxy_redirect off;
    proxy_pass http://unix:/run/gunicorn.sock;
  }
}

```

Enable site and reload:

```

sudo ln -s /etc/nginx/sites-available/django_ninja /etc/nginx/sites-enabled
sudo nginx -t
sudo systemctl restart nginx

```

4.5 HTTPS with Let's Encrypt

Install certbot and request cert:

```

sudo apt install certbot python3-certbot-nginx
sudo certbot --nginx -d yourdomain.com -d www.yourdomain.com

```

Certbot will update your Nginx config to redirect HTTP → HTTPS and manage renewals.

5. Option B: Docker Compose (recommended for reproducible deploys)

Here's a full docker-compose.yml (production-ish, small-scale) that includes Postgres, Redis, web (Gunicorn), worker (Celery), and Nginx.

```

version: '3.8'

services:
  db:
    image: postgres:15
    environment:
      POSTGRES_DB: ninja_db
      POSTGRES_USER: ninja_user

```

```
POSTGRES_PASSWORD: supersecret

volumes:

  - postgres_data:/var/lib/postgresql/data

networks:

  - ninja-net


redis:

image: redis:7

command: redis-server --appendonly yes

volumes:

  - redis_data:/data

networks:

  - ninja-net


web:

build: .

command: gunicorn project.wsgi:application -w 4 -k gthread -b 0.0.0.0:8000

volumes:

  - ./app

env_file:

  - .env.production

depends_on:

  - db

  - redis

networks:

  - ninja-net
```

```
worker:

  build: .

  command: celery -A project worker -l info

  env_file:

    - .env.production

  depends_on:

    - db

    - redis

  networks:

    - ninja-net


nginx:

  image: nginx:stable

  ports:

    - "80:80"

    - "443:443"

  volumes:

    - ./deploy/nginx/conf.d:/etc/nginx/conf.d

    - ./staticfiles:/staticfiles

    - ./mediafiles:/mediafiles

  depends_on:

    - web

  networks:

    - ninja-net
```

```
volumes:

  postgres_data:

  redis_data:


networks:

  ninja-net:
```

Dockerfile (simple):

```
FROM python:3.11-slim

ENV PYTHONDONTWRITEBYTECODE 1

ENV PYTHONUNBUFFERED 1


WORKDIR /app

COPY requirements.txt /app/

RUN pip install --upgrade pip && pip install -r requirements.txt


COPY . /app

RUN adduser --disabled-password djuser

USER djuser


CMD ["gunicorn", "project.wsgi:application", "-w", "4", "-b", "0.0.0.0:8000"]
```

Important files:

- `.env.production` should contain all env vars (`DATABASE_URL`, `SECRET`, `ALLOWED_HOSTS`, etc). Never commit it.
- `deploy/nginx/conf.d/default.conf` should proxy to `web:8000` and serve `/static` from mounted `./staticfiles`.

Commands:

```
# Build & start

docker-compose up -d --build


# Run migrations

docker-compose exec web python manage.py migrate

docker-compose exec web python manage.py collectstatic --noinput


# Create superuser

docker-compose exec web python manage.py createsuperuser
```

This approach is reproducible, easy to scale with multiple web replicas, and portable.

6. Background tasks: Celery + Redis

6.1 Install & configure Celery

```
pip install celery redis

project/celery.py:

import os

from celery import Celery

os.environ.setdefault("DJANGO_SETTINGS_MODULE", "project.settings")

app = Celery("project")

app.config_from_object("django.conf:settings", namespace="CELERY")

app.autodiscover_tasks()

project/__init__.py:

from .celery import app as celery_app

__all__ = ("celery_app",)

Django task example app/tasks.py:

from celery import shared_task
```

```
@shared_task
def add(a, b):
    return a + b
```

Start worker with `celery -A project worker -l info` (Docker Compose example above uses this).

6.2 Beat (periodic tasks)

If you need scheduled jobs:

`celery -A project beat -l info`

Or use flower for monitoring.

7. Scaling strategies

Vertical vs Horizontal

- **Vertical:** bigger machine (more CPU / RAM). Easy, limited.
- **Horizontal:** multiple web worker instances behind a load balancer (recommended for scale).

Scale the web layer

- With Docker Compose, run multiple web containers behind Nginx (or use a dedicated LB).
- In Kubernetes, scale Deployments: `kubectl scale deployment web --replicas=5`.

Database scaling

- Use **replica/read-replicas** for heavy read workloads.
- Use connection pooling (pgbouncer) to reduce DB connection overhead.
- Use managed DB service for automated backups, failover.

Caching

- Use Redis for caching expensive queries, sessions, or rate limiting.
- Example: cache response for 60s:

```
from django.views.decorators.cache import cache_page
```

```
@cache_page(60)

def expensive_view(request):

    ...
```

Horizontal autoscaling (cloud)

- Set autoscaling rules (CPU, request latency) on Kubernetes or cloud service.

8. Kubernetes (brief, for production-grade scaling)

Minimal Deployment (web) YAML (conceptual):

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web
spec:
  replicas: 3
  selector:
    matchLabels:
      app: ninja-web
  template:
    metadata:
      labels:
        app: ninja-web
    spec:
      containers:
        - name: web
          image: yourrepo/ninja:latest
          ports:
```

```

- containerPort: 8000

envFrom:

- secretRef:
    name: ninja-secrets

readinessProbe:

  httpGet:
    path: /health
    port: 8000

livenessProbe:

  httpGet:
    path: /health
    port: 8000

```

Service + Ingress sits in front. Use HorizontalPodAutoscaler for CPU-based autoscaling. Use managed Kubernetes (EKS/GKE/AKS) for easier operations.

9. Migrations & zero-downtime deploys

Safe migration strategy

- Add **new column with nullable default** in code first, deploy, then backfill data, then set NOT NULL in a later deploy.
- Avoid breaking migrations that require heavy locks (e.g., ALTER TABLE on huge tables). Use `pg_repack` or online schema change tools.

Zero-downtime tip

- Use rolling deployments (Kubernetes) or multiple app instances; apply DB migrations on a single maintenance pod/step before switching traffic to new pods; or use feature flags.

10. Health checks & readiness endpoints

Add endpoints that return simple status:

```
from ninja import NinjaAPI
```

```
api = NinjaAPI()

@api.get("/health")
def health(request):
    return {"status": "ok"}
```

Nginx or LB can probe /health for readiness/liveness.

11. Security best practices

- DEBUG=False.
- Use HTTPS (Let's Encrypt / managed TLS).
- Keep SECRET_KEY secret.
- Use strong DB passwords and restrict IP access if possible.
- Use up-to-date packages (dependabot).
- Content Security Policy & rate limiting for public endpoints.
- Sanitize file uploads; store media in private buckets when needed.
- Use CSP, HSTS headers (Nginx config).
- Limit admin access (IP allowlist) and enable 2FA for admin accounts.

Nginx security headers example:

```
add_header X-Frame-Options "SAMEORIGIN";

add_header X-Content-Type-Options "nosniff";

add_header Referrer-Policy "no-referrer-when-downgrade";

add_header Strict-Transport-Security "max-age=31536000; includeSubDomains; preload";
```

12. Backups & Disaster Recovery

Postgres on VPS:

```
# Dump

pg_dump -U ninja_user -h localhost ninja_db > backup_$(date +%F).sql
```

```
# Restore
```

```
psql -U ninja_user -h localhost -d ninja_db < backup.sql
```

For large DBs use `pg_dump -Fc` (custom format) and `pg_restore`.

For managed DBs, use automated daily backups and snapshots.

Store backups offsite (S3) and test restores regularly.

13. Logging, Monitoring & Alerts

- **Logs:** Use structured logging (JSON) and ship to ELK stack or cloud logging (Datadog, CloudWatch).
- **Metrics:** Export metrics (Prometheus) and visualize in Grafana.
- **Uptime:** UptimeRobot or cloud provider health checks.
- **Alerts:** CPU, errors, high latency, DB connection issues should trigger alerts.

Django example logging config (settings.py):

```
LOGGING = {
    "version": 1,
    "disable_existing_loggers": False,
    "handlers": {
        "console": {"class": "logging.StreamHandler"},
    },
    "root": {"handlers": ["console"], "level": "INFO"},
}
```

Consider Sentry for error tracking (sentry-sdk).

14. Performance tuning & common pitfalls

Pitfalls

- N+1 DB queries: use `select_related` / `prefetch_related`.
- Long-running requests: move to background tasks.
- Too many DB connections: use `pgbouncer`.
- Large static/media files served by Django: use Nginx or S3.

- Unmanaged secrets: store in Vault or cloud secret manager.

Tuning

- Increase Gunicorn workers depending on CPU: $\text{workers} \approx 2 \times \text{cores} + 1$ (but test).
- Use async workers (uvicorn) for async endpoints / websockets if needed.
- Cache (Redis) for expensive queries.

15. Example Nginx + Gunicorn production setup (full)

Nginx site (HTTPS handled by certbot):

```
server {  
  
    listen 80;  
  
    server_name example.com;  
  
  
    location /.well-known/acme-challenge/ {  
        root /var/www/certbot;  
    }  
  
  
    location /static/ {  
        alias /srv/django_ninja_app/staticfiles/;  
    }  
  
  
    location /media/ {  
        alias /srv/django_ninja_app/mediafiles/;  
    }  
  
  
    location / {  
        proxy_pass http://unix:/run/gunicorn.sock;
```

```

    proxy_set_header Host $host;

    proxy_set_header X-Real-IP $remote_addr;

    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

    proxy_set_header X-Forwarded-Proto $scheme;

    }
}

```

Gunicorn socket path and systemd file were shown earlier.

16. CI/CD (recommended)

Use GitHub Actions / GitLab CI or cloud CI to:

- Run tests
- Lint code
- Build Docker image
- Push to registry
- Trigger deployment (SSH deploy, Render/GCP deploy, or update Kubernetes manifest)

Example GitHub Actions (sketch):

```

name: CI

on: [push]

jobs:
  test:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3

      - uses: actions/setup-python@v4
        with: {python-version: '3.11'}

      - run: pip install -r requirements.txt

      - run: pytest

```

Add a deployment job after tests pass.

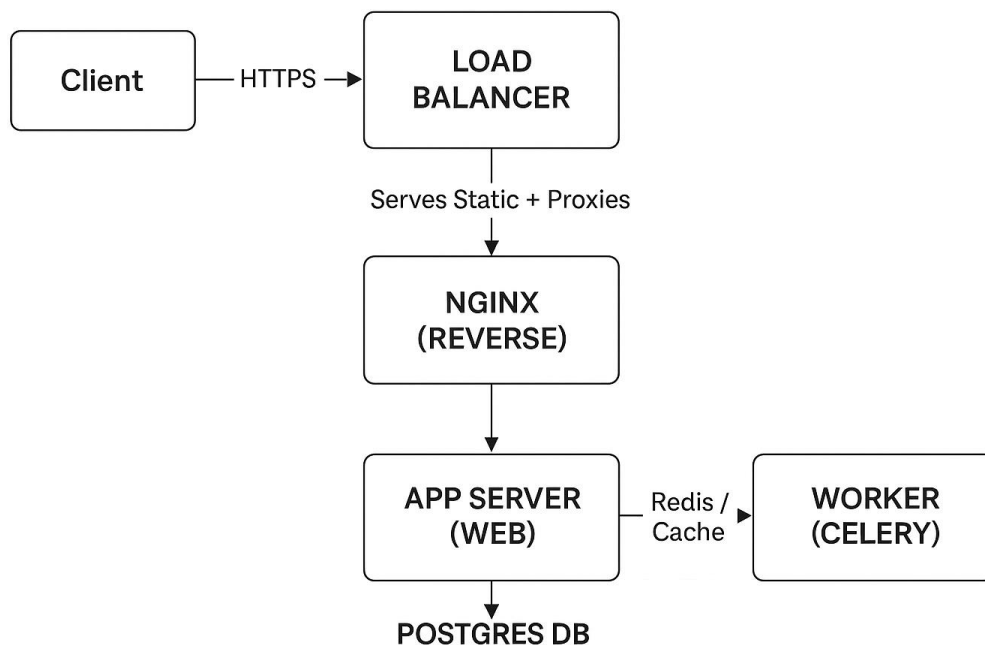
17. Health checks & readiness in Kubernetes / Docker

- Kubernetes uses readiness and liveness probes from earlier.
- For Docker Swarm or simple LB, set an HTTP health check hitting /health.

18. Example: Full deploy sequence (Docker Compose on VPS)

1. SSH to server and install Docker & Docker Compose.
2. Clone repo to /srv/app.
3. Create .env.production with env vars.
4. docker-compose up -d --build
5. docker-compose exec web python manage.py migrate
6. docker-compose exec web python manage.py collectstatic --noinput
7. Configure Nginx to proxy to the Docker Nginx or use the Docker Nginx service.
8. Add certbot for SSL or configure cloud TLS.
9. Run docker-compose ps to confirm services healthy.

19. Small Architecture diagram



20. Troubleshooting quick tips

- 502 Bad Gateway → Check Gunicorn socket path and that Gunicorn is running.
- 500 Internal Server Error → Check Gunicorn logs & Django logs, Sentry for stack trace.
- Static files 404 → Confirm collectstatic ran and Nginx static alias path matches STATIC_ROOT.
- DB connection refused → Verify DB host, user, password, and that firewall allows the connection.
- High latency → Profile DB queries for N+1 or missing indexes.

21. Checklist before hitting “deploy”

- DEBUG=False and YES in env variables
- Secrets in environment, not repo
- Database accessible and migrated
- Static files collected & served via Nginx/CDN/S3
- Gunicorn workers tuned and managed (systemd or containers)
- Nginx configured and TLS set up
- Celery workers and Redis if background tasks used
- Monitoring & logging configured
- Backups tested for DB
- Load test in staging (optional)
- Health endpoints for readiness/liveness

Exercises (hands-on)

1. **Deploy with Docker Compose:** Use the provided docker-compose.yml to deploy locally and test. Run migrations, create a superuser, and visit /api/docs.
2. **Deploy to a small VPS:** Follow the Gunicorn + Nginx + Certbot flow, and serve your API at a real domain.
3. **Add Celery:** Implement a background task (welcome email) and ensure worker picks up the task.
4. **Scale:** Spin up two web replicas behind Nginx and confirm sessionless JWT works across both.
5. **Backup & Restore:** Create a Postgres dump and restore it to a new DB instance.

Quick reference commands

```
# Migrate and static
```

```
python manage.py migrate
```

```
python manage.py collectstatic --noinput

# Gunicorn run (local)

gunicorn project.wsgi:application -w 3 -b 0.0.0.0:8000


# Docker build & up

docker-compose up -d --build


# Create DB dump

pg_dump -U user -h host dbname > backup.sql


# Restore

psql -U user -h host dbname < backup.sql
```

Final words — keep it safe, simple, proven

Deploying is the point where code meets reality. Start small, automate everything you can (CI/CD, infra as code), and monitor closely. Use managed services where they save time (managed Postgres, Load Balancers, Certs). Always test restores and rehearse deployments.

If you want, I can:

- Generate a ready-to-run **Docker Compose** repo scaffold for your project, or
- Produce **Nginx + systemd** config files customized to your project names, or
- Create a **Kubernetes manifest** set and a GitHub Actions pipeline that builds/pushes Docker images and deploys.

Chapter 23

Interview Questions and Best Practices

Stand Out as a Django Ninja and API Engineer

Overview

This chapter includes:

- Core interview questions (with sample answers)
- Conceptual and scenario-based questions
- Best practices for Django Ninja + API development
- Live coding challenges
- Soft skill & collaboration questions
- Tips to stand out as a candidate

21.1: Django Ninja-Specific Questions

These questions focus on the framework and its usage in real-world REST API development.

Q1. What is Django Ninja and how is it different from Django REST Framework (DRF)?

A:

Django Ninja is a modern, fast web framework for building APIs with Django. Key differences from DRF include:

Django Ninja	Django REST Framework
Built on Pydantic for validation	Custom serializers
Faster performance (based on FastAPI principles)	Slower comparatively

Auto docs via OpenAPI + Swagger	Requires config for docs
Simpler routing and schema handling	More flexible but verbose

Q2. What are the key building blocks in a Django Ninja app?

- Router for grouping endpoints
- Schema for validating input and formatting output
- Pydantic models under the hood
- Response type to specify output
- HTTP methods (get, post, put, etc.)

Q3. How do you secure routes in Django Ninja?**A:**

By using security backends like `django_auth` or custom strategies:

```
from ninja import Router

from ninja.security import django_auth

router = Router(auth=django_auth)

@router.get('/secure-data')

def secure_view(request):

    return {"message": "This is secured"}
```

21.2: High-Level Concepts and System Design**Q4. How do you design REST APIs for scalability?**

- Use pagination
- Optimize database queries (`prefetch/select_related`)
- Cache responses where appropriate
- Use async for high IO tasks (if necessary)
- Separate read/write services in large systems (CQRS)
- Document API with Swagger/OpenAPI
- Use load balancing and horizontal scaling

Q5. How would you structure a Django Ninja project for a large system?

Answer may include:

- Domain-driven design structure
- Broken into modules (/api/users, /api/payments, etc.)
- Use services/helpers for business logic outside views
- Use Django signals and middleware smartly
- Use settings for environment-based configs

21.3: Behavioral / Soft Skill Questions

Q: Can you describe a challenge you faced while building a Django API?

Example Answer:

"I was working on a patient management API where some endpoints were very slow. I profiled the API and noticed N+1 query issues, so I used `select_related` and `prefetch_related` to optimize it, reducing the response time by over 70%."

Section 21.4: Live Coding Questions

Turn these into a practice exercise before real interviews.

Task: Create a simple Django Ninja view that filters products based on category and price range.

Expected Solution:

```
@router.get("/products")

def filter_products(request, category: str = None, min_price: float = None, max_price: float = None):

    qs = Product.objects.all()

    if category:

        qs = qs.filter(category__icontains=category)

    if min_price:

        qs = qs.filter(price__gte=min_price)

    if max_price:

        qs = qs.filter(price__lte=max_price)

    return qs
```

21.5: Best Practices

Area	Best Practice
Schemas	Use explicit schemas for input/output
Error Handling	Use global exception handlers
Logging	Log all errors, use correlation IDs for debugging
Security	Use HTTPS, validate data, sanitize inputs
Performance	Avoid N+1, cache when possible
Documentation	Make your /api/docs intuitive and always up to date

21.6: Bonus – What Hiring Managers Look For

- Clean code with clarity and minimalism
- Understanding of design patterns
- Ability to abstract logic and reuse code
- Communication ability
- Demonstrated side projects (like your hospital API)
- Eagerness to learn and collaborate

Final Tip

Always have a mini portfolio ready. A well-structured **GitHub repository** with a Django Ninja project (e.g., the hospital app) gives you an edge. Include a README.md with installation steps, diagrams, and sample API calls.

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Sylvester Benjamin is a Software Engineer and Cybersecurity Consultant, currently serving as the General Manager at Cyber Oracle, a leading firm specializing in software engineering and cybersecurity solutions. With a strong foundation in building secure and scalable systems, he is passionate about integrating cutting-edge technologies with best-practice security measures.

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This book reflects his dedication to bridging the gap between rapid innovation and robust security — ensuring that the tools driving the future are built with safety and clarity at their core.